



Date:

/

Chapter 1: introduction

Chapter goal:

- Get “feel,” “big picture,” introduction to terminology
 - more depth, detail *later* in course

Overview/roadmap:

- What *is* the Internet? What *is* a protocol?
- **Network edge**: hosts, access network, physical media
- **Network core**: packet/circuit switching, internet structure
- **Performance**: loss, delay, throughput
- Protocol layers, service models
- Security
- History

1. What Is "The Internet"?

Internet = **network of networks**

This is the **MOST IMPORTANT** definition.

What Does That Mean?

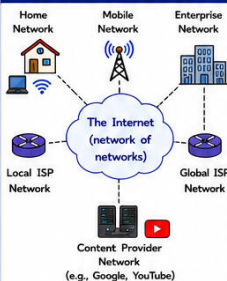
The Internet is NOT one giant network owned by one company. Instead, many smaller networks connect together.

Examples:

- home networks
- school networks
- company networks
- mobile networks
- ISP networks
- datacenter networks

All connected together.

Example From the Picture



Simple Real-Life Analogy

Imagine roads:

- Small streets -----
- City roads -----
- Highways -----
- International roads -----

Connected together.

The Internet works similarly.



2. Hosts (End Systems)

hosts = end systems

What Is a Host?

A host is a device connected to the Internet.

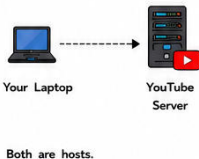
Examples:

- Laptop
- Phone
- Server
- Smart TV
- Gaming Console
- Smart Car

Why "End System"?

Because communication starts or ends there.

Example:



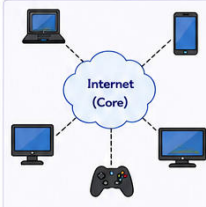
Internet Edge

Applications run at the Internet's **edge**.

Meaning: applications usually run on end devices.

Examples:

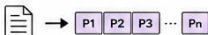
- Browser
- WhatsApp
- Zoom
- YouTube



3. Packet Switches packet switches forward packets

What Is a Packet?

A packet is a small chunk of data. Large messages are split into packets.



What Does "Forward" Mean?

Routers/switches receive packets and send them toward destination. Like postal sorting centers.



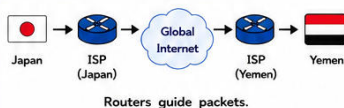
Two Main Packet Switches

Device	Job
Switch	Forwards packets inside a local network (using MAC addresses)
Router	Forwards packets between networks (using IP addresses)

Switch Example (Inside Your Home)



Router Example (Between Countries)



4. Communication Links Physical paths that carry data

Types of Communication Media

Type	Example
Copper	Ethernet cable (copper wires)
Fiber	Optical Internet cables (light pulses)
Radio	WiFi / 4G / 5G (radio waves)
Satellite	Space communication (satellite links)

Bandwidth

Transmission rate = bandwidth

Meaning of Bandwidth

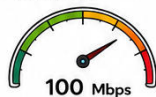
Bandwidth means: how much data can be sent per second.

Usually measured in:

- Mbps (Mega bits per second)
- Gbps (Giga bits per second)

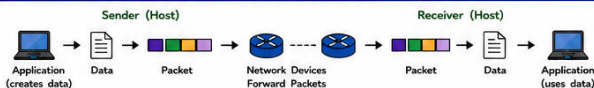
Example

100 Mbps means:



= 100 million bits per second can be sent over the link.

Full Communication Summary



Most Important Concepts

- ✓ Encapsulation: wrapping data with headers
- ✓ Decapsulation: removing headers
- ✓ Message: application-layer unit
- ✓ Segment: transport-layer unit
- ✓ Datagram: network-layer unit
- ✓ Frame: link-layer unit
- ✓ Switch: works mainly at link layer

5. Protocols Are Everywhere

Protocols control sending and receiving messages.

What Is a Protocol?

A protocol is:

- a set of communication rules. Like human languages or traffic rules.



Real-Life Analogy

If two people speak:

- same language
 - same rules
 - they understand each other.
- Computers also need rules.



Examples of Internet Protocols

Protocol		Purpose
HTTP		web pages
TCP		reliable transport
IP		routing packets
WiFi		wireless communication
Ethernet		wired local networks
4G/5G		mobile communication

Example: Loading a Website (google.com)

Layer	Protocol
Application	HTTP
Transport	TCP
Network	IP
Link	WiFi / Ethernet

When opening google.com, many protocols cooperate:

https://www.google.com

6. Internet Standards

RFC

RFC means:
Request For Comments



- These are official Internet technical documents.
- They define protocols.

IETF

IETF means:
Internet Engineering Task Force



- This organization develops Internet standards.

Example

TCP/IP standards are documented by IETF RFCs.



7. Internet Infrastructure

Internet infrastructure provides services to applications.

Meaning

The Internet is the foundation that supports apps.



Applications Using Internet

- Web browsing
- Streaming video
- Email
- Gaming
- Zoom meetings
- Social media
- Cloud apps

Important Idea

Applications do NOT build networks themselves.

They use Internet infrastructure.



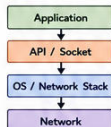
8. Programming Interface

The Internet provides programming interface.

Meaning

Applications use APIs / sockets to communicate.

The OS / network stack provides communication services.



Real-Life Analogy: Postal Service

- You:
- write address
 - send package



Postal system handles:



Similarly:

- Apps send data.
- Internet handles transport.

9. End-to-End Communication Example

Suppose: You watch YouTube on your phone.

Step-by-Step

- Phone creates request
- WiFi sends packets
- Router forwards packets
- ISP forwards packets
- Global Internet routes packets
- YouTube datacenter receives request
- Video packets return back

End-to-End Flow



All automatically!

10. Big Picture of the Entire Slide Set

The Internet consists of:

Component	Role	Example
Hosts (End Systems)	- run applications	
Routers	route packets between networks	
Switches	forward packets in local networks	
Links (Communication Media)	carry data	
Protocols	communication rules	
ISPs (Internet Service Providers)	connect networks	
Datacenters / Servers	provide services and content	

VERY Important Final Concept

The Internet is:
a massive global system where:

- billions of devices
- use shared protocols
- exchange packets
- through interconnected networks
- using layered communication systems.



Key Takeaways

- ★ The Internet = network of networks
- ★ Hosts are end systems where communication starts/ends
- ★ Packet switches (switches & routers) forward packets
- ★ Communication links carry data
- ★ Protocols control communication
- ★ Standards (RFC, IETF) define protocols
- ★ Internet infrastructure supports applications
- ★ APIs / sockets provide programming interface
- ★ End-to-end communication happens automatically

- What *is* the Internet?
- What *is* a protocol?
- **Network edge:** hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- History

The Internet: 3 Main Parts Explained

1. NETWORK EDGE

The outer part of the Internet where users and devices exist.

- hosts: clients and servers
- servers often in data centers

What is a Host?

A host is any device connected to the Internet.



Client (Requests Services)

A client is a device that requests services or data.

Example:
Your browser opening YouTube, your phone opening Instagram.



Server (Provides Services)

A server is a powerful computer that provides services / data.

Example:
YouTube servers, Google servers, Netflix servers.



Data Centers

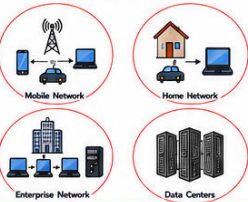
Servers are often located in data centers.



- Thousands of computers
 - Cooling systems
 - Power systems
 - Networking equipment
- Purpose: store websites, apps, cloud services.

Examples: Google Data Centers, Amazon AWS, Microsoft Azure

Edge Examples (Red Circles)



Edge devices either use the Internet or provide Internet services.

Main Devices You Should Remember

Device	Purpose
Client	Requests data from servers
Server	Provides data and services
Router	Routes packets between networks
Switch	Connects devices in the same network
Access Point	Provides wireless connection
Modem	Connects home / office to ISP

2. ACCESS NETWORKS & PHYSICAL MEDIA

How devices connect to the Internet.

Access Network

An access network connects users/devices to the Internet.



Physical Media (Communication Path)

The actual medium that carries data.

Wired Communication

Uses cables.

- Examples:
- Ethernet cable
 - Fiber optic cable



Advantages:

- ✓ Fast
- ✓ Stable
- ✓ Low interference

Wireless Communication

Uses radio waves.

- Examples:
- Wi-Fi
 - Bluetooth
 - Cellular networks



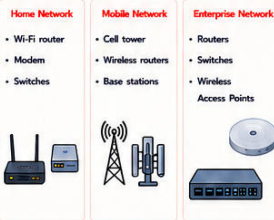
Advantages:

- ✓ Mobility
- ✓ Convenience

Disadvantages:

- ✗ Interference
- ✗ Weaker signals

Access Network Equipment (Red Circles)



These devices move data from users into the Internet.

3. NETWORK CORE

The backbone of the Internet.

- interconnected routers
- network of networks

What is the Network Core?

The network core is the high-speed backbone that moves data between networks worldwide.



Router

A router forwards packets between networks.

It decides: "Which path should this packet take?"



ISP (Internet Service Provider)

ISPs connect users to the Internet.

Examples: AT&T, Verizon, NTT, SoftBank



Local / Regional ISP

Smaller providers. Connect cities or regions to the Internet.



National / Global ISP

Large worldwide backbone providers that connect countries and continents.

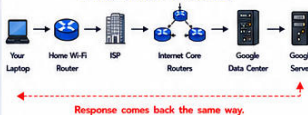


Network of Networks

The Internet is not one single network. It is many networks connected together.

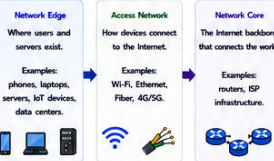


Complete Internet Flow (Example)

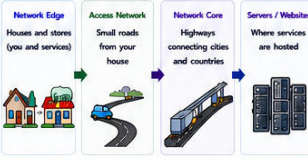


Response comes back the same way.

Big Picture Summary



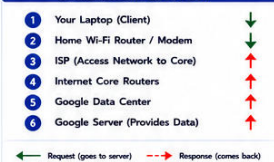
Simple Analogy (Delivery System)



Comparison of the 3 Parts

Part	Main Role	Examples	Key Devices
Network Edge	Where data originates or is used	Phones, laptops, servers, IoT devices, data centers	Clients, Servers
Access Network	How data enters the Internet	Wi-Fi, Ethernet, Fiber, 4G/5G, DSL	Modem, Router, Switch, Access Point
Network Core	Moves data across the Internet	ISP networks, backbone routers	Routers, ISP Infrastructure

Full Internet Flow (Step-by-Step)



Key Takeaways

- ✓ The Internet has 3 main parts: Edge, Access, Core.
 - ✓ Edge = where users and services exist.
 - ✓ Access = how devices connect.
 - ✓ Core = the backbone that moves data worldwide.
 - ✓ Routers, ISPs, and networks work together to deliver information quickly and reliably.
- Think of it like a delivery system:
-

1. ACCESS NETWORKS AND PHYSICAL MEDIA

How do end systems connect to the edge router?

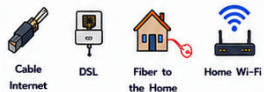
End Systems

Devices that connect to the Internet.



Main Types of Access Networks

1 Residential Access Networks (Used in Homes)



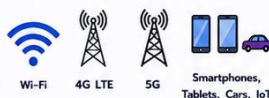
2 Institutional / Enterprise Networks (Used in schools, universities, companies)



- ✓ High speed
- ✓ More reliable
- ✓ Larger scale
- ✓ Advanced infrastructure

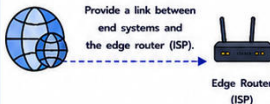
3 Mobile Access Networks (Wireless)

Wireless access using Wi-Fi, 4G, 5G.



Access Network Goal

Provide a link between end systems and the edge router (ISP).



Key Takeaway

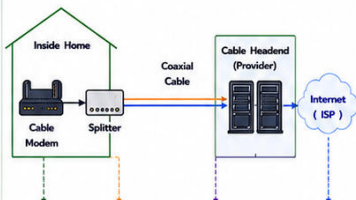
Access networks are the "last mile" that bring the Internet from the ISP to homes, organizations, and mobile users using wired or wireless physical media.



2. CABLE-BASED ACCESS NETWORKS

Internet Access over Cable TV Infrastructure

Main Components



Cable Modem	Splitter	Shared Cable Line	Cable Headend
Installed in your house. Converts cable signals into Internet data.	Splits the signal into: • TV signal • Internet signal	Many homes share the same cable line.	Central location of the cable company. Connects homes to the ISP.

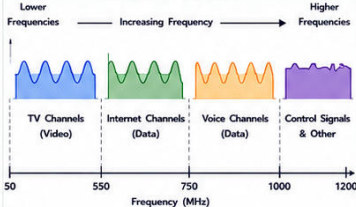
Frequency Division Multiplexing (FDM)

Different channels transmitted in different frequency bands.

- Frequency 1 → TV (Video)
- Frequency 2 → Internet (Data)
- Frequency 3 → Voice (Data)
- Frequency 4 → Control Signals & Other

Different frequencies prevent interference.

Channels on the Cable



Key Points

- ✓ One cable carries many services simultaneously using different frequencies.
- ✓ Homes share the same cable → shared bandwidth.
- ✓ Performance depends on how many users are active at the same time.
- ✓ Cable systems support TV, Internet, Voice, and other services.

Real-Life Example

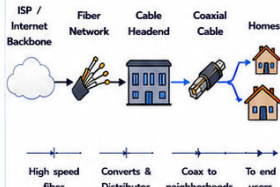
When many people in a neighborhood stream video at the same time, the Internet speed may slow down because they share the same cable network.



3. HFC — HYBRID FIBER COAX

A Combination of Fiber Optic and Coaxial Cable

How HFC Works



High speed fiber → Converts & Distributes → Coax to neighborhoods → To end users

Shared Access Network

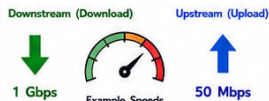


Many homes share the access network. If many users download heavily at the same time, speed can decrease.



Asymmetric Speeds

Downstream (to users) is much faster than upstream (from users). Because most users download more than upload.



Why Asymmetric?

- Most Internet activities are downloads: streaming, browsing, updates.
- Less data is uploaded: messages, photos, emails.
- Allows providers to offer higher download speeds cost-effectively.



Key Advantages of HFC

- ✓ Uses existing cable TV infrastructure.
- ✓ High downstream speeds.
- ✓ Cost-effective for wide coverage.
- ✓ Supports triple-play: TV, Internet, Voice.



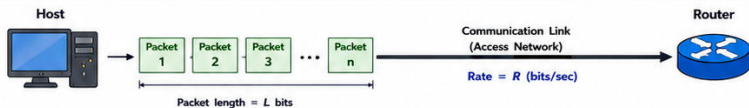
Bottom Line

HFC networks combine the speed of fiber and the reach of coaxial cable to deliver high-speed Internet to many homes efficiently and economically.



Host: Sends Packets of Data

A host (computer) sends data over a network by breaking a large message into packets and transmitting each packet into the access network.



1 Application Message

Takes application message created by applications.



Examples:

- Email text
- Video stream
- Webpage
- Image file
- File download

Example: You want to send a 10 MB file.

2 Breaking into Packets

Breaks the message into smaller chunks, known as packets.

P1 P2 P3 P4 ... Pn

Why use packets?

- ✓ Routers handle small data better
- ✓ Errors affect only one packet
- ✓ Many users can share the network
- ✓ More efficient use of bandwidth

Example: A large file becomes many packets.

3 Packet Length (L)

Each packet has a length of L bits.



L = Packet size (in bits)

A bit is the smallest unit of data: 0 or 1.

Example: If $L = 1000$ bits, one packet has 1000 bits.

4 Transmission Rate (R)

The host transmits the packet into the access network at transmission rate R .



R = Transmission rate (bits per second)
= Link capacity
= Link bandwidth

Example: If $R = 1$ Mbps, 1,000,000 bits are sent every second.

5 Transmission Delay

Time needed to push all packet bits into the link.

$$d_{trans} = \frac{L}{R}$$

Where:

L = packet size (bits)
 R = transmission rate (bits/sec)
 d_{trans} = transmission delay (sec)

Understanding the Formula $d_{trans} = \frac{L}{R}$

Key Ideas

Bigger Packet → More Delay

If L increases, more bits to send → delay increases.



Faster Link → Less Delay

If R increases, link is faster → delay decreases.



Examples

Example 1 (Slow Link)

$L = 1000$ bits
 $R = 100$ bits/sec

$$d_{trans} = \frac{1000}{100} = 10 \text{ sec}$$

The host needs 10 seconds to completely place the packet onto the link.

Example 2 (Fast Link)

$L = 8000$ bits
 $R = 2$ Mbps
= 2,000,000 bits/sec

$$d_{trans} = \frac{8000}{2 \times 10^6} = 0.004 \text{ sec} = 4 \text{ ms}$$

Because the link is very fast, the packet enters quickly.

Example 3 (Large Packet)

$L = 1,000,000$ bits
 $R = 1$ Mbps
= 1,000,000 bits/sec

$$d_{trans} = \frac{1,000,000}{1,000,000} = 1 \text{ sec}$$

Even though the link is fast, a very large packet still takes time.

Example 4 (Multiple Packets)

Host sends 5 packets
 $L = 12000$ bits (each)
 $R = 10$ Mbps
= 10,000,000 bits/sec

Step 1: One packet delay

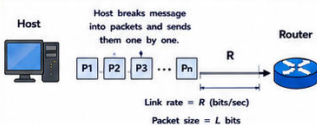
$$d_{trans} = \frac{12000}{10 \times 10^6} = 0.0012 \text{ sec} = 1.2 \text{ ms}$$

Step 2: Five packets

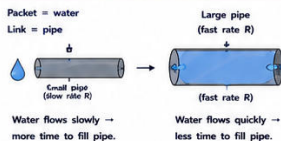
$$5 \times 1.2 \text{ ms} = 6 \text{ ms}$$

Answer: Total = 6 milliseconds

Diagram Explanation



Water Pipe Analogy



Important Clarification

Transmission delay ONLY means: time to place all packet bits onto the link. ✓

It does NOT include:

- ✗ Travel time across the Internet
- ✗ Router processing delays
- ✗ Propagation delay
- ✗ Queuing delays



Summary Table

Concept	Meaning
Host	Computer or device that sends data
Packet	Small chunk of data
L	Packet size (bits)
R	Transmission rate (bits per second)
Transmission Delay (d_{trans})	Time required to push all packet bits onto the link

Most Important Formula

$$d_{trans} = \frac{L}{R}$$

where,

L = packet size (bits)

R = transmission rate (bits/sec)

d_{trans} = transmission delay (seconds)

Quick Reference

- ✓ Increase L → Increase delay
- ✓ Increase R → Decrease delay
- ✓ High-speed links reduce delay
- ✓ Packets make networking efficient
- ✓ Each packet is transmitted separately



Links: Physical Media

When two devices communicate: Computer → cable / wireless → another device
the data must travel through some medium. That medium is called: **Physical Media**.

1 Bit Propagation

Bit propagates between transmitter / receiver pairs.

What is a Bit?
Smallest unit of data:
0 or 1

Transmitter

Sends signals.

Examples:

- laptop
- router
- phone

Receiver

Receives signals.

Examples:

- another computer
- switch
- router

Propagate Means

To travel through the medium.

Examples:

- electrical signal in copper wire
- light in fiber cable
- radio wave in air



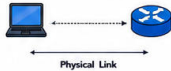
2 Physical Link

What lies between transmitter & receiver.

The actual connection carrying data.

Examples:

- Ethernet cable
- fiber optic cable
- Wi-Fi radio waves



3 Guided Media

Signals propagate in solid media.

Examples:

- copper wire
- fiber optic cable
- coaxial cable

Guided Media Examples

Medium	Signal Type
Copper (Twisted Pair)	Electrical
Fiber Optic	Light
Coaxial Cable	Electrical

Signals are confined inside a physical medium.



4 Unguided Media

Signals propagate freely (no physical wire).

Examples:

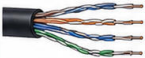
- Wi-Fi
- Bluetooth
- 4G / 5G
- Radio



Signals travel through air / space.

Twisted Pair (TP) Cable

Two insulated copper wires twisted around each other.



Why Twisted?

Reduces electromagnetic interference (EMI).

Used in:

- home Ethernet
- office networks
- LAN connections



Ethernet Categories

Category	Description	Supports
Cat5	Up to 100 MHz	100 Mbps 1 Gbps
Cat6	Up to 250 MHz	10 Gbps

Practical Example (Twisted Pair)

Suppose: Cat6 Ethernet Speed = 10 Gbps

Question: How long to send 1 GB file?

Step 1 – Convert File Size

1 byte = 8 bits
1 GB = 8 Gb



Step 2 – Use Formula

$t = \frac{\text{Data}}{\text{Rate}}$

$t = \frac{8 \text{ Gb}}{10 \text{ Gbps}} = 0.8 \text{ seconds}$

Answer: It takes 0.8 seconds to send 1 GB over Cat6 (10 Gbps).

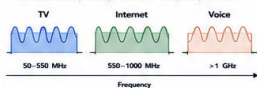
Coaxial Cable

Two concentric copper conductors.
Commonly used for cable TV and cable Internet.



Broadband (Multiple Frequency Channels)

One coax cable can carry many signals simultaneously using different frequency bands.



Practical Example (Coax)

Suppose: 1 channel = 200 Mbps

Question: How much data can be sent in 5 seconds?

Use: $\text{Data} = \text{Rate} \times \text{Time}$

Calculation:

$$200 \text{ Mbps} \times 5 \text{ sec} = 1000 \text{ Mb}$$

Answer: It takes megabits of data can be sent in 5 seconds.

Fiber Optic Cable

Glass fiber carrying light pulses.
Instead of electricity, light carries the data.



Each Pulse Represents a Bit

- Light ON = 1
- Light OFF = 0



Advantages of Fiber



Very High Speed
10s – 100s Gbps
Extremely fast.



Low Error Rate
Resistant to electromagnetic noise.

Practical Example (Fiber)

Suppose: Fiber speed = 100 Gbps

Movie size = 50 Gb

Question: Transmission time?

$t = \frac{\text{Data}}{\text{Rate}}$

$$t = \frac{50 \text{ Gb}}{100 \text{ Gbps}} = 0.5 \text{ seconds}$$

Answer: It takes only 0.5 seconds to send a 50 Gb movie over 100 Gbps fiber.

Summary Table

Concept	Meaning
Host / Device	Computer, phone, server that sends / receives data
Bit	Smallest unit of data (0 or 1)
Packet	Small chunk of data for transmission
Link (Physical)	Actual path between transmitter and receiver
Guided Media	Signals travel inside a solid medium (wire, fiber)
Unguided Media	Signals travel freely through air / space
Transmission Delay	Time to push all packet bits onto the link (L / R)

Key Formulas

Transmission Delay

$$d_{\text{trans}} = \frac{L}{R}$$

Where:

L = Packet size (bits)

R = Transmission rate (bits/sec)

d_{trans} = Transmission delay (seconds)

Quick Comparison

Medium	Signal Type	Speed	Noise Immunity	Use Cases
Twisted Pair (TP)	Electrical	Up to 10 Gbps	Low	LAN, Ethernet
Coaxial Cable	Electrical (RF)	100 Mbps – 1+ Gbps	Medium	Cable TV, Internet
Fiber Optic	Light	10s – 100s Gbps	Very High	Backbone, Long distance
Wireless (Wi-Fi, 4G/5G)	Radio Waves	Up to few Gbps	Low (interference)	Mobility, Wireless access

Wireless Radio

No physical wire. Signals travel through air using **electromagnetic waves**.

1. Wireless Radio Basics

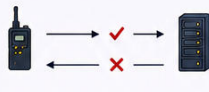
Data is carried by radio waves that propagate through the air.



- No physical wire or cable.
- Uses electromagnetic waves to transmit data.

Half-Duplex

Devices cannot transmit and receive simultaneously on the same channel.



2. Wireless Problems

Wireless signals can be affected by the environment.

Reflection

Signals bounce off walls, buildings, or other surfaces.



Obstruction

Objects block or weaken signals.

Examples: walls, trees, people.



Interference / Noise

Other signals disturb communication.

Examples: nearby Wi-Fi, microwaves, Bluetooth, radio signals.



3. Radio Link Types

Different wireless technologies are used for various applications.

1. Wireless LAN (Wi-Fi)

- Typical speeds: 10 – 100s Mbps
- Range: tens of meters



Practical Example (Wi-Fi)

Suppose: Wi-Fi speed = 100 Mbps

Video size = 500 Mb

Time needed?

$$t = \frac{500}{100} = 5 \text{ seconds}$$

2. Wide-Area Cellular (4G/5G)

- Used for smartphones and mobile Internet.
- Coverage: kilometers



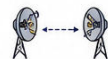
3. Bluetooth

- Short-distance communication.
- Examples: earbuds, keyboards, smart watches.



4. Terrestrial Microwave

- Point-to-point wireless links.
- Used between towers.



5. Satellite

- Internet from satellites in space.
- Example: Starlink



Satellite Delay

Signals travel very long distances (to space and back), causing high delay.
Typical round-trip delay: ~ 270 ms

Practical Example (Satellite)

Suppose: Round-trip delay = 270 ms

How many seconds?

$$270 \text{ ms} = 0.27 \text{ sec}$$

4. Details of Radio Link Types

Type	Description	Typical Speed	Range / Coverage	Use Cases
Wireless LAN (Wi-Fi)	Local area wireless network inside homes, offices, buildings.	10 – 100s Mbps	Tens of meters	Home Internet, office networks, laptops, phones
Wide-Area Cellular (4G/5G)	Mobile broadband over cellular towers.	10 Mbps – 1 Gbps+	Kilometers (city-wide)	Smartphones, tablets, mobile Internet
Bluetooth	Short-range wireless connection.	1 – 3 Mbps	Up to 10 meters	Earbuds, keyboards, wearables, IoT
Terrestrial Microwave	High-frequency point-to-point links between towers or buildings.	100 Mbps – Gbps	Tens of km (line of sight)	Backhaul links, ISP connections
Satellite	Data transmitted to/from satellites in space.	10 Mbps – 1 Gbps+	Global (very wide)	Rural Internet, maritime, aviation, Starlink

5. Guided vs Unguided Summary

Guided Media (Wired)

Signals travel through physical cables.



- Examples: Ethernet, fiber optic, coaxial
- More stable, less affected by environment
- Usually higher security

Unguided Media (Wireless)

Signals travel through air (using radio waves).



- Examples: Wi-Fi, 4G/5G, Bluetooth, Satellite
- Flexible, supports mobility
- More affected by interference and noise

6. Satellite Delay Explanation



Radio waves travel at the speed of light
($\approx 3 \times 10^8$ m/s).

Round-trip delay ≈ 270 ms

Practical Example

$$270 \text{ ms} = 0.27 \text{ seconds}$$

7. Important Media Comparison

Medium	Signal Type	Typical Speed	Advantages	Disadvantages	Typical Use
Twisted Pair (Cat5/Cat6/Cat6a)	Electrical	100 Mbps – 10 Gbps	- Cheap - Easy to install - Widely available	- Susceptible to interference - Shorter distance	LAN networks, office wiring, home Ethernet
Coaxial Cable	Electrical (RF)	100 Mbps – 1+ Gbps	- Broadband (multiple channels) - Good for long distances	- Shared medium - Bulkier	Cable TV, Cable Internet
Fiber Optic Cable	Light	10s – 100s Gbps	- Extremely high speed - Low attenuation - Very low error rate	- Expensive equipment - Fragile	Backbone networks, Data centers, Long-distance links
Wireless Radio	Radio Waves	10 Mbps – Gbps+	- Mobility and flexibility - Easy deployment	- Interference / noise - Lower security - Variable performance	Wi-Fi, 5G, Bluetooth, Satellite, Microwave

8. Key Takeaways

- Wireless radio uses electromagnetic waves instead of cables.
- Half-duplex means no simultaneous send and receive.
- Wireless faces reflection, obstruction, and interference.
- Different radio technologies are used based on range and speed needs.
- Satellite links have very high delay.
- Choose the right medium based on speed, cost, range, and reliability.



Date: /

Chapter 1: roadmap

- What is the Internet?
- What is a protocol?
- Network edge: hosts, access network, physical media
- **Network core:** packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models
- History



The Network Core

The network core is the heart of the Internet, made of many interconnected routers. Its job is to move packets from the source to the destination.

1. Mesh of Interconnected Routers

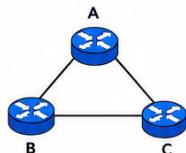
What is a Router?

A router is a networking device that forwards packets between networks.



Mesh Means

Routers are connected together in many ways.

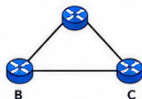


Example: Router A ↔ Router B, Router B ↔ Router C, Router A ↔ Router C

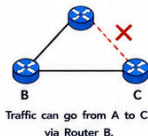
Why Multiple Paths?

If one path fails, another path can be used. This makes the Internet reliable and robust.

Working Normally



If One Link Fails



2. Packet Switching

What is Packet Switching?

- Hosts break application-layer messages into packets.
- Each packet travels independently through the network.

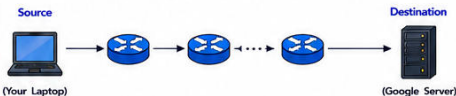
Example

Suppose you send a 20 MB video. The network breaks it into:



How Packets Move

The network forwards packets from one router to the next.



Key Points

- ✓ Packets may take different paths through the network.
- ✓ Routers make forwarding decisions based on the best available path.
- ✓ Packets may arrive at the destination out of order and are reassembled.

Source and Destination

Source (Sender)



Example: Your laptop

Destination (Receiver)



Example: Google server

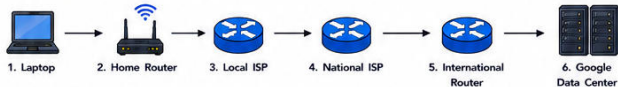
Real Example of Packet Movement

Scenario

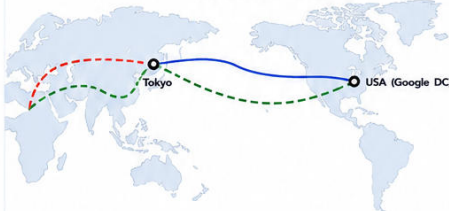
You are in Tokyo and want to watch a YouTube video hosted on a server in the USA.



Possible Route



How Packets Are Delivered



What Happens at Each Router?

- Router receives a packet.
- Checks the destination address.
- Chooses the best next hop.
- Forwards the packet.
- Next router repeats the process.

Important Idea

Different packets from the same message may take different paths through the Internet. They may experience different delays but still reach the destination.



Practical Example — Packet Switching

Suppose:

- A file = 12,000 bits
- The host breaks it into 3 packets

Question: What is the packet size?

Solution

$$\text{Packet Size} = \frac{\text{Total Data}}{\text{Number of Packets}}$$

$$= \frac{12,000 \text{ bits}}{3}$$

Answer

Packet Size = 4,000 bits per packet

Summary

Concept	Meaning
Network Core	The backbone of the Internet.
Router	Forwards packets between networks.
Mesh Topology	Many interconnected routers, multiple paths.
Packet Switching	Data is split into packets and sent independently.
Reliability	If one path fails, packets can use another path.
Source / Destination	Sending and receiving end systems.

Two Key Network- Core Functions

Routers in the network core perform TWO fundamental functions:

- 1) **Forwarding** (local action) and
- 2) **Routing** (global action)

1. FORWARDING (Local Action)

What Does Forwarding Mean?

Forwarding means: move a packet from the input link to the correct output link.

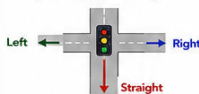


Simple Explanation

- Router receives a packet.
- Then it decides:

"Which port should I send this packet to?"

Like a Road Intersection
Cars arrive at an intersection.
Traffic signs decide where to go: left, right, or straight.



Forwarding Table

The forwarding table tells the router where packets should be sent.

Destination (Header Value)	Output Link
0100	Link 3
0101	Link 2
1001	Link 1
1100	Link 3

Practical Example — Forwarding
Suppose router receives a packet with Destination = 0101.
From the forwarding table:

0101 → Link 2

So, the router sends the packet to Output Link 2.

How Forwarding Works (One Step)



1. Router reads the destination address in the packet header.
2. It looks up the forwarding table.
3. It sends the packet to the selected output link.

2. ROUTING (Global Action)

What is Routing?

Routing means finding the complete path from source to destination across the entire network.



Difference Between Routing and Forwarding

Function	Meaning
Routing (Global)	Chooses the entire path from source to destination.
Forwarding (Local)	Moves the packet one step: input link → output link.

Example

Suppose you travel from Tokyo → Osaka.

- **Routing:**
Google Maps chooses the best highway / path.
- **Forwarding:**
At each intersection, you turn left / right / straight.



Routing Algorithms

Routers use routing algorithms (mathematical methods) to find efficient paths.

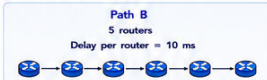
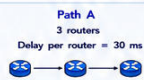
They consider:

- Distance
- Congestion
- Link speed
- Failures
- Load



Practical Example — Routing

Suppose there are two possible paths from Source to Destination.



Which path is faster?

Path A Total Delay
 $3 \times 30 \text{ ms} = 90 \text{ ms}$

Path B Total Delay
 $5 \times 10 \text{ ms} = 50 \text{ ms}$

Answer:
Path B is faster even though it has more routers.

Packet Header

Every packet has a header that contains important information.



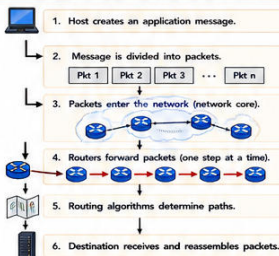
Header Fields (Examples)

- Source Address
- Destination Address ← Used by routers
- Protocol Number
- Sequence Number
- Other Control Information

Routers read the destination address to make forwarding decisions.

Dest: 0101 →

Big Picture: How It All Works



Key Concepts Summary

Concept	Meaning
Network Core	Interconnected routers forming the core of the Internet.
Packet Switching	Data is broken into packets and sent independently.
Router	A device that forwards packets between networks.
Forwarding	Local action: move a packet from input link to correct output link.
Routing	Global action: choose the complete path from source to destination.
Packet Header	Contains destination address and other control information.

Most Important Idea

Routing determines WHERE packets should go.
It is a global decision (across the whole network).



Forwarding determines HOW packets move at each router.
It is a local decision (one step at a time).

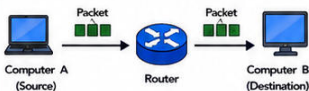


Packet Switching: Store-and-Forward, Queueing, and Packet Loss

1. PACKET SWITCHING — STORE AND FORWARD

Basic Idea

A router must receive the **ENTIRE** packet before forwarding it to the next link.



Step-by-Step Process

- Source sends packet to router.
- Router stores the whole packet in memory.
- After full reception, router forwards packet to next device.

Why Called

Store-and-Forward?

- Because the router:
- stores the packet
 - then forwards it completely.

Transmission Delay Formula

$$d_{trans} = \frac{L}{R}$$

Where:

L = packet size (bits)

R = transmission rate (bits/sec)

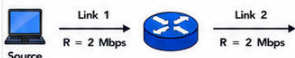
One-Hop Numerical Example

Given: $L = 10$ Kbits, $R = 100$ Mbps

- | 1. Convert Units | 2. Apply Formula | 3. Result |
|---------------------------------|--|------------|
| 10 Kbits = 10,000 bits | $d_{trans} = \frac{10,000}{100 \times 10^6}$ | 0.0001 sec |
| 100 Mbps = 100,000,000 bits/sec | $= 0.0001$ sec | $= 0.1$ ms |

Two Links (Store-and-Forward at Each Router)

Example: $L = 8000$ bits, $R = 2$ Mbps, across 2 links



One-Link Delay

$$d = \frac{8000}{2 \times 10^6} = 0.004 \text{ sec}$$

Two Links (Total)

$$2 \times 0.004 = 0.008 \text{ sec} = 8 \text{ ms}$$

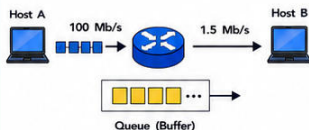
Big Idea

Store-and-forward causes delay at every router because the router must receive the entire packet before forwarding.

2. PACKET SWITCHING — QUEUEING

What is Queueing?

When packets arrive at a router faster than they can be transmitted, they must wait in line (in the router buffer).



Explanation

- Incoming packets arrive at 100 Mb/s.
- Outgoing link can transmit only at 1.5 Mb/s.
- Packets accumulate in the router buffer (queue).

Real-Life Analogy

100 cars arrive per minute at a toll booth, but the booth can allow only 10 cars per minute. Cars pile up → same as packets in a queue.



Practical Example — Queueing

- Incoming traffic = 10 Mbps, Outgoing link = 2 Mbps
- Packets arrive 5 times faster than they can leave.
- Queue grows rapidly.

Queueing Delay Example

Each packet transmission takes 2 ms on the outgoing link.

There are already 5 packets waiting in the queue. How long must a new arriving packet wait?

$$\text{Waiting time} = 5 \text{ packets} \times 2 \text{ ms} = 10 \text{ ms}$$

Answer: 10 ms queueing delay (before transmission begins).

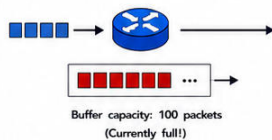
Key Point

Total delay = Queueing delay + Transmission delay

3. PACKET LOSS

What Happens if the Queue Becomes Full?

Routers have limited buffer (memory).



If Buffer Fills Up

- New arriving packets cannot enter the buffer.
- The router drops (discards) them.
- This is called packet loss.



Packet Loss Example

Router buffer capacity = 50 packets
Already in buffer = 50 packets (full)

A new packet arrives → packet is discarded (lost).



Result of Packet Loss

Packet loss can cause:

- Slow downloads
- Video buffering
- Lag in online games
- Retransmissions (more delay)



Important Rule

If Arrival Rate > Transmission Rate for too long:
Queue grows → Buffer fills → Packet loss occurs.

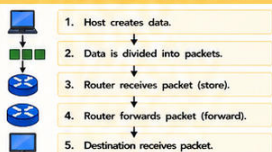


Sustained overload leads to packet loss and poor network performance.

Quick Summary of the Three Concepts

Concept	What It Means	Key Takeaway
1. Store-and-Forward	Router must receive the ENTIRE packet before forwarding it to the next link.	Causes transmission delay at every router.
2. Queueing	Packets wait in line in the router buffer when arrivals are faster than transmission.	Creates queueing delay.
3. Packet Loss	If buffer is full, new packets are dropped.	Leads to retransmissions, slower performance, and poor user experience.

Flow of a Packet in the Network



4. Alternative to Packet Switching — Circuit Switching

What is Circuit Switching?

A dedicated communication path (circuit) is reserved between the source and destination before communication begins.



Used in:
Traditional telephone systems.

Dedicated Circuit on Each Link

Each physical link has multiple circuits (channels). One call gets its own reserved circuit end-to-end.

Example: Each link has 4 circuits



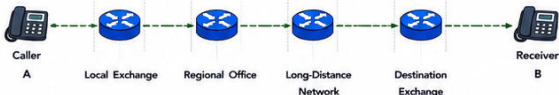
No Sharing

Unlike packet switching, resources are NOT shared between different users.

Once a circuit is established, the full bandwidth is reserved for that call, even if the user is silent.

Main Idea

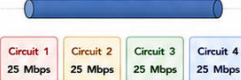
Resources (bandwidth) are reserved in advance for a communication session. The path remains dedicated for the entire duration (no sharing with others).



Bandwidth Allocation Example

Suppose a link has capacity = 100 Mbps and 4 circuits are established on it.

100 Mbps Link Capacity



Calculation

Each circuit gets = $100 \text{ Mbps} / 4 = 25 \text{ Mbps}$

Important Observation

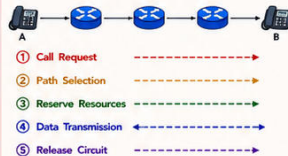
Even if a user is idle and not sending data, the reserved bandwidth remains unused. Other users cannot use it.

Circuit Establishment (Call Setup)

Before communication starts:

1. A path is selected from source to destination.
2. Resources (bandwidth) are reserved on every link.
3. Circuit is established.
4. Data (voice) flows over the dedicated circuit.
5. Circuit is released when call ends.

Example: Call Setup



Circuit Switching vs Packet Switching

Feature	Packet Switching	Circuit Switching
Resource Sharing	Yes (shared)	No (dedicated)
Efficiency	High (good for bursty traffic)	Lower (wasteful if idle)
Delay	Variable (due to queueing)	Predictable (no queueing)
Queueing	Yes	No
Packet Loss	Possible (if buffer overflows)	Very rare (if circuit established)
Setup Delay	None (send immediately)	Yes (circuit setup time)
Used In	Internet (data networks)	Traditional telephone systems

Real-Life Analogy

Packet Switching

Like cars sharing public roads.



Many cars share the road. Sometimes wait in traffic.

Circuit Switching

Like having your own private lane.



You have a private lane. Always available for you.

Practical Example — Circuit Switching

Given:

- Link capacity = 100 Mbps
- 4 circuits established
- You make a phone call

$$\text{Bandwidth per circuit} = \frac{100 \text{ Mbps}}{4} = 25 \text{ Mbps}$$

If you are silent (not talking):

- Your 25 Mbps is still reserved
- Others cannot use it
- Wasted bandwidth occurs

In packet switching, this bandwidth could be used by others.

Advantages of Circuit Switching

- ✓ Predictable delay (no queueing)
- ✓ Guaranteed bandwidth
- ✓ Suitable for constant bit rate traffic (like voice calls)
- ✓ Simple once circuit is established

Good for real-time communication where delay must be minimal.

Disadvantages of Circuit Switching

- ✗ Inefficient if user is idle
- ✗ Setup time before communication starts
- ✗ Not scalable for bursty data
- ✗ Wastes network resources

Not efficient for Internet data traffic, which is bursty (on-off).

Big Picture Summary

Two Main Networking Methods

Packet Switching

Data is divided into packets. Resources are shared. Used in the Internet.



Circuit Switching

A dedicated path is reserved. Resources are not shared. Used in traditional phone systems.



Key Concepts Summary

Concept	Meaning
Store-and-Forward	Router must receive the entire packet before forwarding.
Queueing	Packets wait in line in router buffer when arrival rate > transmission rate.
Queueing Delay	Time packets spend waiting in the queue.
Packet Loss	Packets are dropped when router buffer becomes full.
Circuit Switching	Dedicated path is reserved for entire communication session.
Packet Switching	Resources are shared, packets forwarded hop-by-hop.

Circuit Switching: FDM and TDM

Big Idea

Suppose 4 users want to communicate using one physical link.

Question: How can all users share the same link?

Two solutions:

1. Divide by frequency → FDM
2. Divide by time → TDM

Users



One Physical Link
(Shared by All Users)

Network
(Core)

1. FREQUENCY DIVISION MULTIPLEXING (FDM)

Main Idea of FDM

The communication channel is divided into multiple frequency ranges.
Each user gets its own frequency band.

Analogy

Think of a highway.
Instead of sharing one lane, each user gets a dedicated lane.

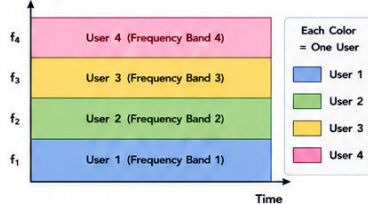


Important Observation

In FDM:

- ✓ All users transmit simultaneously
- BUT:**
- ✓ On different frequencies.

Frequency



All users transmit at the same time (simultaneously)
but on different frequency bands.

Real Examples of FDM

- Radio broadcasting (FM/AM)
- Cable TV
- Older telephone systems

How FDM Works

The total bandwidth of the link is split into several frequency bands.
Each band is separated from others to avoid interference.



Practical Examples — FDM

Example 1: Bandwidth per User

Suppose total bandwidth of the link = 40 MHz.
4 users share the link equally.

Question: How much bandwidth does each user get?

Solution:

$$\begin{aligned}\text{Bandwidth per user} &= \frac{\text{Total Bandwidth}}{\text{Number of Users}} \\ &= \frac{40 \text{ MHz}}{4} \\ &= 10 \text{ MHz per user}\end{aligned}$$

Answer

Each user gets
10 MHz

Example 2: Total Required Bandwidth

Suppose each user needs 5 Mbps.
There are 6 users.

Question: What is the total bandwidth required?

Solution:

$$\begin{aligned}\text{Total Bandwidth} &= \text{Number of Users} \times \text{Bandwidth per user} \\ &= 6 \times 5 \text{ Mbps} \\ &= 30 \text{ Mbps}\end{aligned}$$

Answer

Total bandwidth
required is
30 Mbps

Advantages of FDM

- ✓ All users can communicate simultaneously.
- ✓ Continuous transmission (no waiting time).
- ✓ Efficient for analog signals.
- ✓ Well-suited for broadcasting (radio, TV).

Disadvantages of FDM

- ✗ Wasted bandwidth if some users are idle.
- ✗ Requires guard bands between channels (wastes some bandwidth).
- ✗ Interference can occur between frequency bands.
- ✗ Not very efficient for digital data with bursty traffic.

2. TIME DIVISION MULTIPLEXING (TDM) — Preview

Main Idea of TDM

The link is divided into time slots.
Each user gets the whole bandwidth
but only for a short time.
Users take turns.

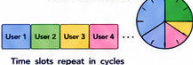


Diagram Explanation



Users share the entire bandwidth,
but each user transmits in its own time slot.

Real Examples of TDM

- Digital telephone systems
- ISDN
- Satellite communication
- Cellular networks (GSM)

FDM vs TDM Comparison

Feature	FDM (Frequency Division Multiplexing)	TDM (Time Division Multiplexing)
Division Basis	Frequency bands	Time slots
Transmission	All users transmit at the same time	Users transmit in turns
Bandwidth per User	Always allocated (even if idle)	Used only in assigned time slot (efficient)
Best For	Analog signals, broadcasting	Digital data, bursty traffic
Efficiency	Lower	Higher

Key Takeaway



FDM divides by frequency.

TDM divides by time.

Both allow multiple users to share the same physical link in circuit switching.

2. TIME DIVISION MULTIPLEXING (TDM)

Main Idea of TDM

Instead of dividing frequencies, divide **time**.
Each user gets specific **time slots**.

Analogy

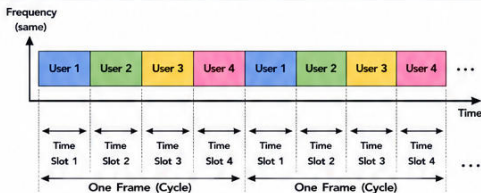
Imagine 4 people sharing one microphone.
Each person speaks during their assigned time.

Important Observation

In TDM:

- ✓ Only **ONE** user transmits at a time.
- ✗ Users take turns.

Diagram Explanation (Same Frequency)



Users share the entire bandwidth, but each user transmits in its own time slot.

Real Examples of TDM

- Digital telephony (e.g., PCM)
- GSM (mobile systems)
- ISDN
- Some satellite systems

Practical Example 1 — Effective Speed per User

Suppose link speed = 100 Mbps.
4 users share equally.

Question: What is the effective average speed per user?

Solution:

$$\begin{aligned}\text{Effective speed per user} &= \frac{\text{Total Link Speed}}{\text{Number of Users}} \\ &= \frac{100 \text{ Mbps}}{4} \\ &= 25 \text{ Mbps}\end{aligned}$$

Answer

Each user gets 25 Mbps (on average)

Practical Example 2 — Time Slot per User

Suppose time frame (cycle) = 20 ms.
4 users exist.

Question: How long is the time slot per user?

Solution:

$$\begin{aligned}\text{Time slot per user} &= \frac{\text{Total Frame Time}}{\text{Number of Users}} \\ &= \frac{20 \text{ ms}}{4} \\ &= 5 \text{ ms}\end{aligned}$$

Answer

Each user transmits for 5 milliseconds per cycle.



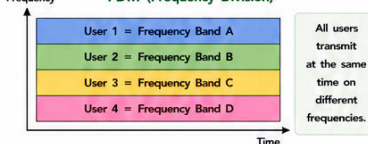
During its time slot, a user can use the **FULL** link speed (100 Mbps in the example).
But it can transmit only during its own slot, not continuously.

FDM vs TDM — Comparison

Feature	FDM (Frequency Division Multiplexing)	TDM (Time Division Multiplexing)
Division Type	Frequency	Time
Users Transmit	Simultaneously (at the same time)	One at a time (take turns)
Resource Allocated	Frequency bands	Time slots
Efficiency	Lower (wasted if some users idle)	Higher (better utilization)
Delay	Lower delay	Waiting delay possible (until next slot)
Best For	Broadcasting, cable TV, analog signals	Digital telephony, data networks
Example	Cable TV channels	Digital telephone calls (PCM)

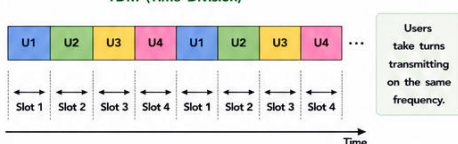
Visual Understanding

FDM (Frequency Division)



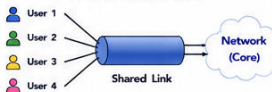
All users transmit at the same time on different frequencies.

TDM (Time Division)



Big Picture

Both FDM and TDM are ways to share one communication link between multiple users.



Goal: Efficiently use the link and support multiple users.

Connection to Circuit Switching

In circuit switching, resources are reserved for the entire call/session.



- FDM reserves frequency bands (dedicated spectrum).
- TDM reserves time slots (dedicated time intervals).

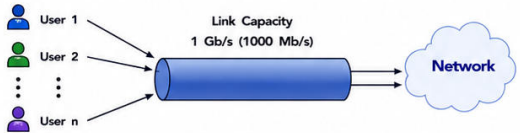
Key Concepts Summary

Concept	Meaning
FDM	Divide the link by frequency. Each user gets a frequency band.
TDM	Divide the link by time. Each user gets a time slot.
Frequency Band	A portion of the total frequency spectrum assigned to a user.
Time Slot	A short time interval assigned to a user.
Circuit Switching	Dedicated path with reserved resources (frequency or time).

Packet Switching vs Circuit Switching

Given:

- Link speed = 1 Gb/s = **1000 Mb/s**
- Each active user needs = **100 Mb/s**
- Each user active = **10%** of time



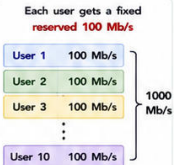
1. CIRCUIT SWITCHING

Bandwidth is reserved for each user, even when the user is idle.

Each user needs Link total bandwidth
100 Mb/s 1000 Mb/s

$$\text{Max users} = \frac{\text{Link bandwidth}}{\text{Bandwidth per user}}$$

$$= \frac{1000}{100} = \mathbf{10 \text{ users}}$$



Answer:

Circuit switching = 10 users

Because each user gets a fixed reserved 100 Mb/s.

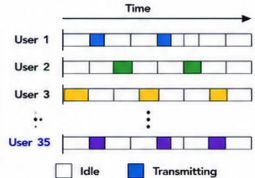
2. PACKET SWITCHING

Users share the link.
Each user is active only 10% of the time.
So not all users transmit at the same time.

The slide says:

35 users

Small risk that more than 10 users become active at the same time.



Answer:

Packet switching = 35 users

But there is a small risk that more than 10 users become active at the same time.

3. WHY MORE THAN 10 ACTIVE USERS IS BAD?

The link can support only 10 active users:

$$10 \times 100 \text{ Mb/s} = 1000 \text{ Mb/s} \quad (\text{OK})$$

If 11 users are active:

$$11 \times 100 \text{ Mb/s} = 1100 \text{ Mb/s} \quad (\text{Exceeds link!})$$

This causes:

- Queueing happens
- Packet delay increases
- Packets may be lost



4. HOW DID THEY GET 0.0004?

Let:

X = number of active users at the same time
 $n = 35$ users, $p = 0.1$ (10% active)

So, $X \sim \text{Binomial}(35, 0.1)$

We want: $P(X > 10)$

Probability that more than 10 users are active

Formula:

$$P(X > 10) = 1 - \sum_{k=0}^{10} \binom{35}{k} (0.1)^k (0.9)^{35-k}$$

Result:

$$P(X > 10) \approx \mathbf{0.0004}$$

So the chance of overload is very small:

$$\mathbf{0.0004 = 0.04\%}$$

5. SIMPLE MEANING

With packet switching:

35 users can share the link

because most users are inactive most of the time.

But sometimes more than 10 users may be active together.

That probability is only about:

$$\mathbf{0.0004 (0.04\%)}$$

Very small.

6. FINAL COMPARISON

Type	Users Supported	Why?
Circuit Switching	10 users	Fixed bandwidth reserved for each user.
Packet Switching	35 users	Users share link statistically.

7. KEY TAKEAWAYS

- ✓ Circuit switching reserves bandwidth → inefficient when users are idle.
- ✓ Packet switching uses bandwidth only when users are active.
- ✓ Packet switching supports many more users with small risk of congestion.
- ✓ This is why the Internet uses packet switching!

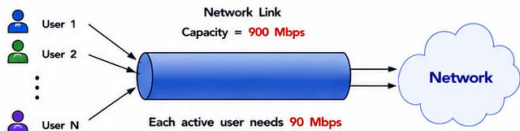


Big Idea: Circuit switching guarantees resources. Packet switching shares resources efficiently. More users, but with a small chance of delay/loss.

Circuit Switching vs Packet Switching – Problems and Solutions

GIVEN

- Network capacity = **900 Mbps**
- Each active user needs = **90 Mbps**
- Each user active = **20%** of time



1. How many users can the network support using circuit switching?

In circuit switching, bandwidth is reserved for each user.

Why?

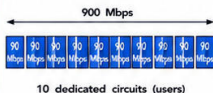
Because:
 $10 \times 90 \text{ Mbps} = 900 \text{ Mbps}$
 The whole link becomes fully reserved.

Calculation:

$$900 \text{ Mbps} \div 90 \text{ Mbps} = 10$$

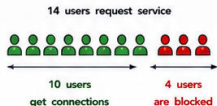
Answer:

10 users



2. If 14 users request service, what happens?

In circuit switching:
 Bandwidth is reserved permanently.
 The link supports only 10 dedicated circuits (users) because $900 / 90 = 10$.



Answer:

B. Only 10 users can connect

No dynamic sharing happens in circuit switching.

3. In packet switching, if there are 20 users, how many are active on average?

Each user active probability = $20\% = 0.2$

Bandwidth needed:

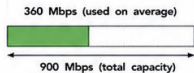
Expected active users:

$$20 \times 0.2 = 4 \text{ users}$$

$$4 \times 90 \text{ Mbps} = 360 \text{ Mbps}$$

Meaning:

Usually only 4 users transmit at the same time.



$360 \text{ Mbps} < 900 \text{ Mbps} \rightarrow$ Works efficiently!

4. If there are 40 users, how many are active on average?

Each user active probability = $20\% = 0.2$

Meaning:

Usually only 8 users transmit at the same time.

Expected active users:

$$40 \times 0.2 = 8 \text{ users}$$

Bandwidth needed:

$$8 \times 90 \text{ Mbps} = 720 \text{ Mbps}$$

$$720 \text{ Mbps (used on average)}$$

900 Mbps (total capacity)

$720 \text{ Mbps} < 900 \text{ Mbps} \rightarrow$ Normally still OK!

5. Why can packet switching support more users?

- Packet switching uses higher bandwidth
- Most users are inactive most of the time
- Packet switching reduces packet size
- Circuit switching is slower

Answer:

B. Most users are inactive most of the time

Explanation:

- ✓ Packet switching works because not all users transmit simultaneously.
- ✓ Users share the link dynamically.
- ✓ If one user is inactive, their bandwidth can be used by others.
- ✓ This is called **statistical multiplexing**.

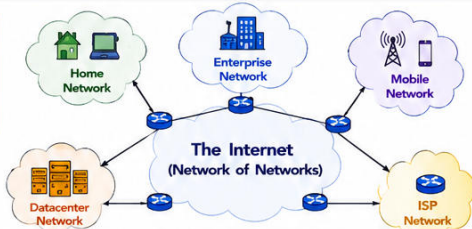


FINAL SUMMARY

Topic	Result / Explanation
Circuit switching users	10 users (because $900 / 90 = 10$. Bandwidth is reserved.)
14 users requesting service (circuit switching)	Only 10 users can connect; 4 users are blocked.
Packet switching with 20 users	4 users active on average ($20 \times 0.2 = 4$). Bandwidth used $\approx 360 \text{ Mbps}$.
Packet switching with 40 users	8 users active on average ($40 \times 0.2 = 8$). Bandwidth used $\approx 720 \text{ Mbps}$.
Why packet switching supports more users	Because most users are inactive most of the time (statistical multiplexing).

Internet Structure: "Network of Networks"

- The Internet is **NOT** one giant single network.
- It is many smaller networks connected together.
- That is why we call it: **"a network of networks"**.



1. Hosts Connect Through ISPs

A device (host) like a laptop, phone, or server does not connect directly to the whole Internet.

Instead, it connects to an ISP (Internet Service Provider).

Examples of ISPs

- Home Wi-Fi company
- Mobile carrier
- University network

Example Path



2. Different Types of Networks

Network Type	Meaning
Home network	Wi-Fi at home
Enterprise network	Company or university network
Mobile network	Cellular network (4G/5G, etc.)
Datacenter network	Servers storing websites/apps
ISP network	Connects users to the Internet

All these networks are interconnected.
Together they form the Internet (a network of networks).

3. Why ISPs Must Connect Together

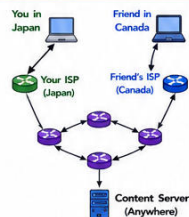
Suppose you are in Japan and your friend is in Canada.

For packets to travel:

your ISP must connect to other ISPs until packets reach the destination.

Goal:

Any two hosts anywhere can communicate.

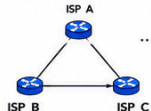


4. Problem: Millions of ISPs

There are huge numbers of ISPs worldwide.

Question: Should every ISP connect directly to every other ISP?

That would look like:



With many ISPs:

- ISP A ↔ ISP B
- ISP A ↔ ISP C
- ISP B ↔ ISP C
- ...

This creates enormous numbers of connections.

5. Why Direct Connection Does NOT Scale

The number of direct connections grows as $O(N^2)$.

If there are N ISPs, possible direct links:

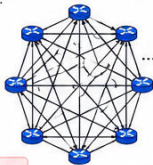
$$\frac{N(N-1)}{2}$$

Example:

If there are 1000 ISPs:

$$\frac{1000 \times 999}{2} = 499,500 \text{ links}$$

Too expensive, too complex, and not scalable!

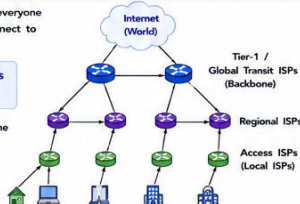


6. Solution: Global Transit ISPs

Instead of connecting everyone directly, small ISPs connect to large ISPs called:

Global Transit ISPs (Tier-1 ISPs)

These are giant backbone networks that connect the world.



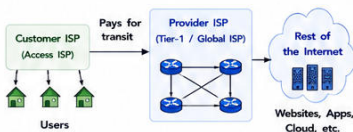
7. Customer-Provider Relationship

Smaller ISPs (customers) connect to larger ISPs (providers) and pay for Internet transit.

Example:

Home ISP pays Tier-1 ISP to carry Internet traffic.

This is an economic agreement.



Key Takeaway

- ✓ The Internet is a network of networks.
- ✓ Hosts connect through ISPs.
- ✓ ISPs must connect to other ISPs.
- ✓ Direct connections between all ISPs are not scalable ($O(N^2)$).
- ✓ Global Transit ISPs (Tier-1) solve the scalability problem.
- ✓ Smaller ISPs pay larger ISPs for transit.

8. Competing ISPs

One giant ISP is unrealistic.

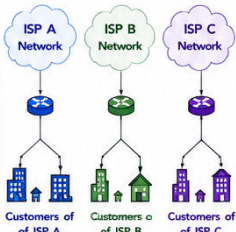
Why?

Because other companies compete.

So we have:

- ISP A
- ISP B
- ISP C

Each has its own customers and network.



9. Internet Exchange Point (IXP)

Different ISPs still need communication. So they meet at:

IXP = Internet Exchange Point

What happens at an IXP?

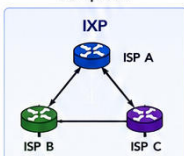
ISPs exchange traffic directly.

This is called: **Peering**.

Benefits:

- ✓ Faster communication
- ✓ Lower cost
- ✓ Less congestion

Example IXP



Peering Link

A peering link is a direct connection between ISPs.
Example: ISP A ↔ ISP B without using another provider.

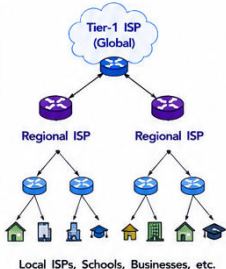
10. Regional ISPs

Regional ISPs connect:

- Small local ISPs
- Schools
- Businesses

to larger Internet providers.

They act as a middle layer.



11. Content Provider Networks

Big companies like:

- Google
- Microsoft
- Amazon
- Netflix

often build their own networks.

Why?

To move content closer to users.



Example

When watching YouTube:



video may come from a nearby Google server. Instead of traveling across the world.

This improves:

- ✓ Speed
- ✓ Latency
- ✓ Reliability

12. Tier-1 ISPs

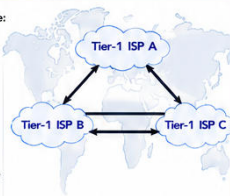
At the center of the Internet are:

Tier-1 ISPs

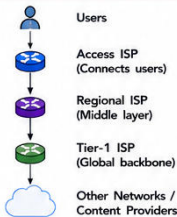
These are extremely large global networks.

They:

- ✓ Connect countries
- ✓ Own backbone infrastructure
- ✓ Exchange traffic globally



Overall Internet Hierarchy



Key Idea

The Internet works because many independent networks cooperate through:

- ✓ Routing
- ✓ Peering
- ✓ IXPs
- ✓ Transit agreements

Important Terms Summary

Term	Meaning
ISP	Internet Service Provider. A company that provides Internet connectivity.
Access ISP	Connects end users (home, mobile, business) to the Internet.
Regional ISP	Connects local ISPs, schools, and businesses to larger Internet providers.
Tier-1 ISP	Large global backbone providers that interconnect the world.
IXP (Internet Exchange Point)	A physical location where ISPs exchange traffic directly.
Peering	Direct traffic exchange between ISPs at an IXP (or private link).
Content Provider Network	Private network built by companies (e.g., Google, Netflix) to deliver content.

Simple Real-World Analogy: Internet Like Airlines

Internet	Airline Analogy
Home / Access ISP	Local Airport
Regional ISP	National Airport
Tier-1 ISP	International Hub
IXP	Airport Transfer Hub



Just like passengers travel through multiple airports and connections, packets travel through many networks and exchange points until they reach the destination.



The Internet is a network of networks, built through cooperation, competition, and smart interconnection!

Chapter 1: roadmap

-
- A night view of a stone bridge over a river in Paris, with the Eiffel Tower visible in the background. The bridge has multiple arches and is illuminated. The river reflects the lights. The background shows the city skyline with the Eiffel Tower prominent.

Packet Delay: Four Sources

- When a packet travels across a network, it does **NOT** move instantly.
- At every router/network device, delays happen.
- Total delay is called: **nodal delay**

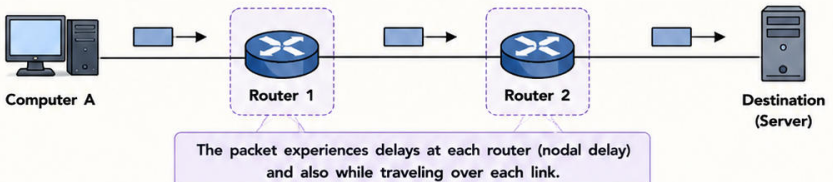
The slide gives:

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

This means total delay has four components:

- 1 Processing delay
- 2 Queueing delay
- 3 Transmission delay
- 4 Propagation delay

Overall Journey of a Packet



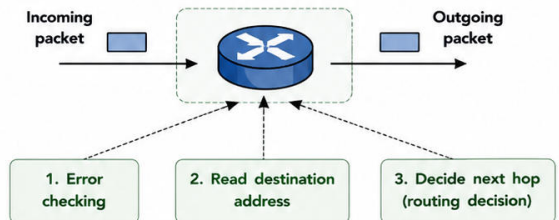
1. Processing Delay (d_{proc})

This is:
time needed for the router to examine the packet.

The router must:

- check for errors
- read destination address
- decide where to send the packet next

What happens inside the router?



Example

Router checks:

"Should this packet go left or right?"



This usually takes:

microseconds

Very small delay.

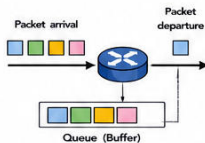
2. Queueing Delay (d_{queue})

Packets may wait in line before transmission.

Like cars waiting at traffic lights.

Why Queueing Happens

- Many packets may arrive together.
- But the router can send only limited packets at a time.
- So packets wait in a queue (buffer).



Important Point

Queueing delay depends on: **network congestion**

- | | |
|--|--|
| If traffic is light:
✓ small queueing delay | If traffic is heavy:
✗ large queueing delay |
|--|--|

Real-Life Analogy

Networking	Real Life
Packet queue	Cars in traffic
Router	Toll gate
Congestion	Traffic jam



3. Transmission Delay (d_{trans})

This is: time required to push all packet bits onto the link.

NOT travel time.

Only: time to place bits onto the wire.

Formula

$$d_{trans} = \frac{L}{R}$$

Where:

Symbol	Meaning
L	Packet size (bits)
R	Transmission rate (bps)

Example

Suppose:

- packet size = 1000 bits
- link speed = 1 Mbps

Then:

$$d_{trans} = \frac{1000}{10^6} = 0.001 \text{ sec}$$

Transmission delay = 1 ms



Key Idea

- Larger packet → larger transmission delay
- Higher link speed → smaller transmission delay

4. Propagation Delay (d_{prop})

This is: time for signal to physically travel through the medium.

Now the bits are already on the wire.

They still need time to travel.

Formula

$$d_{prop} = \frac{d}{s}$$

Where:

Symbol	Meaning
d	Distance
s	Signal propagation speed

Example

Suppose:

- distance = 2000 km = 2×10^6 m
- speed = 2×10^8 m/s

$$d_{prop} = \frac{2 \times 10^6}{2 \times 10^8} = 0.01 \text{ sec}$$

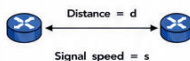
Propagation delay = 10 ms

Propagation Speed

Usually approximately:

$$s \approx 2 \times 10^8 \text{ m/s}$$

which is near the speed of light in cables.



VERY Important Difference

Students often confuse:

Transmission Delay

Concerned with:
How fast bits are pushed out

Depends on:

- distance (L)
- link speed (R)

vs.

Propagation Delay

Concerned with:
How far the signal travels

Depends on:

- distance (d)
- signal speed (s)

Simple Analogy: Water Pipe

Transmission Delay

Like:
time needed to pour all water into the pipe.



Propagation Delay

Like:
time for water to reach the other side.



Complete Delay

Total delay:

$$d_{total} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

Which Delay Changes Most?

Delay	Usually
Processing delay (d_{proc})	Very small (microseconds)
Transmission delay (d_{trans})	Depends on packet size and link speed
Propagation delay (d_{prop})	Depends on distance
Queueing delay (d_{queue})	MOST variable

Why Queueing Is Important

Queueing causes:

- Internet slowdown
- Lag
- Congestion
- Packet loss

Especially during:

- gaming
- video streaming
- downloads
- online classes
- live meetings

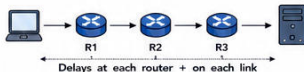
Summary Table

Delay Type	Meaning	Formula	Depends On
Processing delay (d_{proc})	Router examines packet (check errors, read address, make decision)	— (very small)	Router hardware & software
Queueing delay (d_{queue})	Packet waits in buffer (queue) before transmission	— (varies)	Traffic load (congestion)
Transmission delay (d_{trans})	Time to push all packet bits onto the link	$\frac{L}{R}$	Packet size (L), link speed (R)
Propagation delay (d_{prop})	Time for signal to travel through the medium	$\frac{d}{s}$	Distance (d), signal speed (s)



Key Takeaway

Every packet experiences four delays at every router. Queueing delay is the most unpredictable and has the biggest impact on performance!



1. Packet Queuing Delay (Revisited)

Packets sometimes wait in routers before they can be transmitted.
When traffic is high, queues build up → delay increases → congestion occurs!

Important Variables

Symbol	Meaning
λ	Average packet arrival rate (packets per second)
L	Packet size (bits)
R	Link bandwidth / transmission rate (bits per second)

Traffic Intensity (Key Measure)

Traffic intensity ρ (rho) is:

$$\rho = \frac{\lambda L}{R}$$

- λ = offered load (bits per second)
- R = link transmission rate (bits per second)
- ρ = traffic intensity (dimensionless)

ρ compares:

How fast packets arrive VS How fast the router can send them

Why This Matters?

- If packets arrive slowly → small queueing delay.
- If packets arrive faster → queue builds.
- If packets arrive too fast → **queue grows without bound!**

★ Queueing delay is the most unpredictable delay in networks because it depends on traffic conditions, not fixed parameters.

Simple Interpretation of Traffic Intensity ($\rho = \lambda L / R$)

1) $\rho \approx 0$ (Very Light Traffic)

- Packets arrive slowly.
- Router easily handles them.
- Queue is almost always empty.

Result: Small queueing delay

Like:  Empty highway



2) $\rho \rightarrow 1$ (Heavy Traffic)

- Packets arrive almost as fast as the router can transmit.
- Queue builds up.
- Packets wait longer and longer.

Result: Large queueing delay

Like:  Busy road with traffic



3) $\rho > 1$ (Overloaded / Congestion)

- Packets arrive **FASTER** than the router can process.
- Queue grows without bound.
- Average delay becomes theoretically infinite.

Result: Extreme congestion (System unstable)

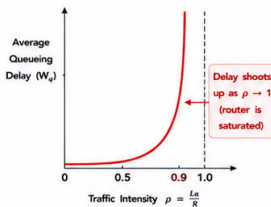
Like:  Traffic jam - no movement!



Why the Graph Suddenly Rises?

This graph shows average queueing delay (W_q) versus traffic intensity ($\rho = \lambda L / R$).

- ★ As ρ approaches 1, the delay increases **VERY rapidly!**
- The router is becoming overloaded.



Real-Life Analogy: Supermarket Checkout

Cashier = Router

Can process: 10 customers per minute



If 2 customers/min arrive



No line. Fast service.

Good!

If 10 customers/min arrive



Line begins forming.

Delays grow

If 20 customers/min arrive



Line grows endlessly.

Congestion disaster!

Key Takeaway

- ✓ Queueing delay depends on traffic intensity ($\rho = \lambda L / R$).
- ✓ As ρ increases, delay increases slowly at first, then shoots up near $\rho = 1$.
- ✓ If $\rho > 1$, the system is unstable → infinite delay!
- ✓ Managing traffic (keeping ρ well below 1) is critical for good network performance.



Summary Table

Traffic Intensity ($\rho = \lambda L / R$)	Situation	What Happens	Average Queueing Delay
$\rho \approx 0$	Light Traffic	Packets arrive slowly. Queue is almost empty.	Small (near 0)
$\rho \rightarrow 1$	Heavy Traffic	Packets arrive as fast as the router can transmit. Queue builds up.	Large (increases rapidly)
$\rho > 1$	Congestion (Overloaded)	Packets arrive faster than router can process. Queue grows forever.	Infinite (theoretically)

★ Keep traffic intensity comfortably below 1 for low delays and good network performance!

2. "Real" Internet Delays and Routes — Traceroute

1. What Is Traceroute?

Traceroute is a network tool that discovers the **path** packets take to reach a destination and measures **delay** at each step.



- Shows the route (hop by hop)
- Shows routers along the path
- Shows delay to each router

2. Real Internet Path (Example)

When you visit a website, your packet passes through many routers.



Traceroute discovers this path and measures delay at each hop.

3. Main Idea of Traceroute

Traceroute sends packets step-by-step through the network. It measures how long each step takes.

TTL (Time To Live)

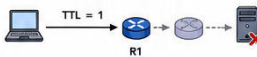
Traceroute uses a field in IP packets called **TTL = Time To Live**.

What TTL Does

- Each router decreases TTL by 1.
- When TTL becomes 0, the router discards the packet and sends a reply back to the sender.

4. How Traceroute Works (Step by Step)

Step 1: TTL = 1



- Send packet with TTL = 1
- Packet reaches first router (R1).
- TTL becomes 0.
- R1 discards packet and sends reply.

Now sender learns:

- First router (R1)
- First-hop delay

Step 2: TTL = 2



- Send packet with TTL = 2
- Packet reaches R1 (TTL=1), then R2 (TTL=0).
- R2 discards packet and sends reply.

Now sender learns:

- Second router (R2)
- Second-hop delay

Step 3: TTL = 3, 4, 5, ...



- Keep increasing TTL: 3, 4, 5, ...
- Each time, the next router replies.
- Continue until the final destination is reached.

Now sender learns:

- Next routers in path
- Delay to each hop

5. Why 3 Probes?

Traceroute sends 3 packets (probes) per hop because:

- Delays vary.
- Some packets may be lost.
- Multiple probes help compute average delay (more accurate).



3 probes to each hop

6. What Traceroute Shows (Typical Output)

```

$ traceroute example.com
traceroute to example.com (93.184.216.34), 30 hops max
Hop  Router (IP)              3 Probe Delays (ms)    Avg (ms)
 1  192.168.1.1 (Home Router)   0.8   1.2   1.1   1.0
 2  10.0.0.1 (ISP Router)       4.3   5.1   4.7   4.7
 3  203.0.113.1 (Regional ISP) 19.3  21.0  19.6  20.0
 4  198.51.100.5 (Intl Router) 98.0  101.6 100.4 100.0
 5  93.184.216.34 (example.com) 118.2 120.1 119.0 119.1
    
```

Hop = step number | Router = router name (or IP) | Delay = round-trip time (ms)

7. Why Traceroute Is Useful

Helps diagnose:

- ✓ Slow Internet
- ✓ Congestion
- ✓ Packet loss
- ✓ Routing problems

Example:

If one router shows **500 ms delay** while others are normal, that router/link is likely congested.

8. Connection to Previous Slide

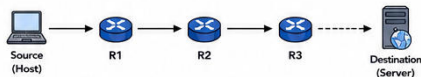
Traceroute reveals **REAL** delays caused by four sources at each router:

- Processing delay**
Router examines the packet
- Queueing delay**
Packet may wait in buffer
- Transmission delay**
Time to push bits onto the link
- Propagation delay**
Time for signal to travel on link

Traceroute measures the total effect of all these delays on the path.

9. Complete Big Picture

Packet travels:



At every router, these things happen:

1 Processing

Router processes the packet

2 Queueing

Packet may wait in the queue

3 Transmission

Bits are put onto the link

4 Propagation

Signal travels on the link

Traceroute sends packets with increasing TTL values. Routers reply when TTL reaches 0. We learn each hop and measure the delay to that hop.

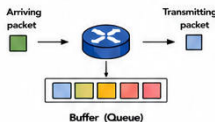
10. Summary Table

Traffic Intensity (λ_a / R)	What It Means	Situation	Queueing Delay
$\lambda_a / R \approx 0$	Packets arrive much slower than router can send	Light traffic	Small
$\lambda_a / R \rightarrow 1$	Packets arrive almost as fast as router can send	Heavy traffic	Large (delays grow quickly)
$\lambda_a / R > 1$	Packets arrive faster than router can send	Congestion (Overloaded)	Infinite growth (theoretically)

Packet Loss

1. What is a Buffer?

A router has a temporary storage area called: **buffer** (also called queue).



Why Buffer Exists

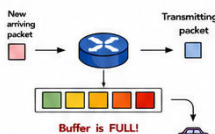
- Packets may arrive faster than the router can send them.
- So packets wait inside the router.
- Like cars waiting in traffic.

2. Packet Loss in Action

The router is currently:

- ✓ Transmitting one packet
- ✓ Storing several packets in buffer

Another packet arrives...



Result:

- The new packet cannot enter.
- It is dropped (discarded).
- This is called: **Packet Loss**.

3. Simple Analogy

Imagine a parking lot with limited spaces.

Cars = packets

Parking spaces = buffer slots

If the parking lot is full:

- New cars cannot enter.
- They leave.



Same idea in networking.

4. Why Packet Loss Happens

Main reason:

network congestion.

Too many packets arrive at the same time.
Router cannot handle all of them.

Relation to Queueing Delay

Recall traffic intensity:

$$\rho = \frac{\lambda a}{R}$$

When ρ becomes very high:

1. Queue grows
2. Delay increases
3. Eventually buffer becomes full

Then: **packet loss** starts.

5. What Happens After Packet Loss?

Option 1 — Retransmission (Reliable)

The sender notices missing packet and sends again.

Common in: TCP

Examples:

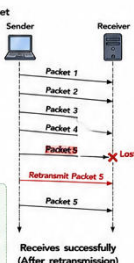
- Web browsing
- File downloads
- Email

Goal: Reliable communication.

Example

If packet #5 disappears:

1. Sender retransmits packet #5
2. Until receiver gets it.



Option 2 — No Retransmission (Fast but Unreliable)

Some applications prefer speed over reliability.

Common in: UDP

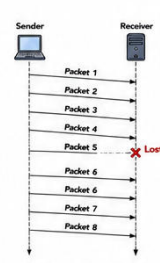
Examples:

- Live video
- Online games
- Voice calls

Goal: Low delay, real-time performance.

Example

If packet #5 disappears: It is not sent again. Receiver continues with remaining packets.



Missing packet tolerated

6. Why Real-Time Apps Avoid Retransmission

In video calls or live streams, if a packet arrives too late, retransmitting may be useless. Better to continue smoothly than to delay more.

7. Packet Loss Effects

Packet loss may cause:

Application	Effect
Video streaming	Blurry video
Gaming	Lag / stutter
Voice calls	Broken / choppy sound
Downloads	Slower transfer

8. Important Insight

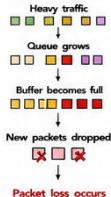
Packet loss is NOT usually caused by physical destruction of packet.

Instead:

Router intentionally drops packets because buffer is full.



9. Complete Flow (Congestion → Loss)



10. Difference Between Delay and Loss

Concept	Meaning
Queueing delay (Delay)	Packet waits longer in the queue.
Packet loss (Loss)	Packet is discarded and never delivered.

Key Difference	
Delay	→ packet still delivered (Just late)
Loss	→ packet never delivered

11. Real Internet Example

Suppose your Wi-Fi router can handle 100 packets/sec.

But suddenly 500 packets/sec arrive. Buffer fills quickly. Extra packets are dropped.

You experience:

- Lag
- Buffering
- Slow Internet



12. Important Networking Terms

Term	Meaning
Buffer	Temporary storage for packets in router
Queue	Waiting line of packets in buffer
Congestion	Too much traffic in the network
Packet loss	Packet dropped due to full buffer
Retransmission	Sending a lost packet again

13. Big Picture: What Routers Do

Packet travels:



At every router:

1. Processing
Router examines the packet

2. Queueing
Packets may wait in buffer

3. Transmission
Bits are put onto the link

4. Propagation
Signal travels along the link

Traceroute helps us measure the total delay at each hop (router).



When traffic is low → small delay, no loss.

When traffic is high → queues grow → buffer fills → packets are dropped → packet loss!

Managing congestion is the key to fast, reliable networks.



Throughput

1. What is Throughput?

- Throughput means:
How fast data successfully travels from sender to receiver.
- Measured in:
bits per second (bps)
- Examples:
 - 1 Mbps
 - 100 Mbps
 - 1 Gbps

Simple Meaning

If a network sends 1 million bits every second, then throughput is **1 Mbps**.

2. Water Pipe Analogy

Think of network links like water pipes.

	Water	= bits (data)
	Pipe width	= bandwidth (capacity)
	Flow speed	= throughput

 Narrow pipe → less water can flow.
Narrow link → less data can flow.

 Wide pipe → more water can flow.
Wide link → more data can flow.

3. Two Types of Throughput

1. Instantaneous Throughput

Speed at ONE exact moment.

Example:

- Current download speed = 80 Mbps right now.



80 Mbps

2. Average Throughput

Average speed over a longer time.

Example:

A 1 GB file downloaded in 100 seconds.

Average throughput =

$$\frac{1 \text{ GB}}{100 \text{ s}}$$

4. Throughput in a Simple Network Path



Important Idea:

The entire connection speed (throughput) depends on the **slowest link**.

5. Bottleneck Link

- The slowest link that limits throughput is called the **bottleneck link**.
- Like a traffic jam on a narrow road.
- Even if all other roads are fast...
ONE narrow road slows everything down!

Bottleneck (narrow road)



6. Case 1: $R_s < R_c$ (Sender Link is Slower)



Example:

- Sender link $R_s = 10$ Mbps
- Receiver link $R_c = 100$ Mbps

Maximum throughput = 10 Mbps
(Limited by sender link)

Bottleneck link:
 R_s

7. Case 2: $R_s > R_c$ (Receiver Link is Slower)



Example:

- Sender link $R_s = 100$ Mbps
- Receiver link $R_c = 10$ Mbps

Maximum throughput = 10 Mbps
(Limited by receiver link)

Bottleneck link:
 R_c

8. Core Rule

$$\text{End-to-end throughput} = \min(R_s, R_c)$$

Smallest rate wins.

The bottleneck link determines the actual throughput.

9. Real-Life Example

Suppose you download a file from the Internet.

Link / Component	Speed (Capacity)	Notes
Home Wi-Fi (to ISP)	50 Mbps	Fast
ISP / Fiber Backbone	1 Gbps	Very fast
Server Upload Speed	20 Mbps	Slowest!

Actual throughput = 20 Mbps
(Limited by server upload speed)

Because server is the bottleneck.

10. Summary Table

Situation	Relationship	Bottleneck	Throughput
Sender link is slower	$R_s < R_c$	Sender link (R_s)	$= R_s$
Receiver link is slower	$R_s > R_c$	Receiver link (R_c)	$= R_c$
Both equal	$R_s = R_c$	Both links	$= R_s = R_c$

11. Key Takeaways

- ✓ Throughput is how fast data successfully reaches the receiver.
- ✓ Measured in bits per second (bps).
- ✓ There are instantaneous and average throughput.
- ✓ Throughput is limited by the bottleneck (slowest link).
- ✓ End-to-end throughput = $\min(R_s, R_c)$.



Big Picture: Data travels through many links. The slowest link sets the speed of the entire journey!



3. Network Scenario — Many Users Share Same Backbone Link

Now MANY users share the same backbone link.

The middle link (backbone) has capacity: R

There are **10** simultaneous connections sharing it.

Senders (Users)



Sender link
capacity = R_s

Backbone link
capacity = R

Receivers



Receiver link
capacity = R_c

All 10 connections pass through the same backbone link (capacity = R).
So they share this link fairly.

Shared Bandwidth

If backbone speed is:

$$R = 100 \text{ Mbps}$$

and 10 users share fairly:

Each gets approximately:

$$\frac{R}{10}$$

So: **10 Mbps each**

Per-Connection Throughput

Per-connection throughput:

$$\min\left(R_c, R_s, \frac{R}{10}\right)$$

Meaning: throughput depends on the smallest among:

- Receiver capacity (R_c)
- Sender capacity (R_s)
- Shared backbone portion ($\frac{R}{10}$)

Example

Suppose:

Component	Notation	Speed
Sender link capacity	R_s	50 Mbps
Receiver link capacity	R_c	30 Mbps
Shared backbone (per user)	$\frac{R}{10}$	5 Mbps

Actual throughput = $\min(50, 30, 5) = 5 \text{ Mbps}$
because shared backbone is the bottleneck.

Important Networking Insight



High-speed Internet alone does **NOT** guarantee fast downloads.

Throughput may be limited by:

- ✓ Server speed
- ✓ Congestion on the network
- ✓ Shared backbone bandwidth
- ✓ Wi-Fi quality / signal
- ✓ ISP limits / policies



Throughput depends on the **bottleneck link!**

Difference Between Bandwidth and Throughput

Concept	Meaning
Bandwidth	Maximum possible capacity of a link (theoretical limit)
Throughput	Actual achieved data rate (real-world speed)



Bandwidth is how much a link **CAN** handle.
Throughput is how much it actually **DOES** handle.

Example

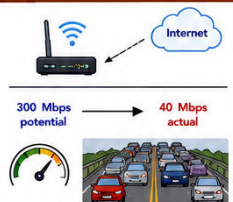
Your Wi-Fi may support:

300 Mbps
bandwidth

But actual throughput may be:

40 Mbps
only

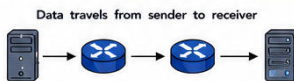
due to congestion or weak signal.



Big Picture

Throughput answers:

How much useful data
is really delivered per second?



but every connection is limited by the **SLOWEST** link in its path.

So the actual throughput of each connection is:

$$\min\left(R_c, R_s, \frac{R}{10}\right)$$

Controlled by the **bottleneck link**.

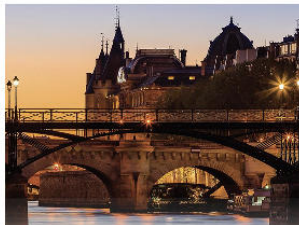


Throughput is not about the fastest link — it's about the **slowest link** on the path!



Find the bottleneck, improve the throughput!

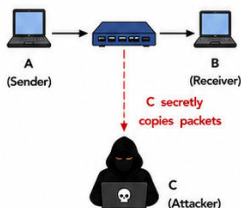
- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- **Security**
- Protocol layers, service models
- History



1. Packet Interception (Packet Sniffing)

Packet sniffing is an attack in which an attacker secretly listens to network traffic and captures packets.

What Is Happening?



Inside a Packet

src: B	Source address
dest: A	Destination address
payload	Actual data

Payload may contain:

- Passwords
- Messages / Emails
- Credit card data (if not encrypted)

Broadcast Media (Shared Environment)

In shared Ethernet, wireless networks, many devices share the same medium.

Examples:

- Public Wi-Fi
- Coffee shop Wi-Fi
- Old Ethernet hubs

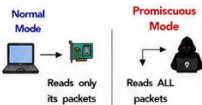


All traffic passes through the same air/cable, so attacker can try reading it.

Promiscuous Mode

Normally, a network card reads ONLY packets meant for it.

In **promiscuous mode**, the network interface reads ALL packets passing by.



Real Example

On a public Wi-Fi, if a website uses plain HTTP (not HTTPS), an attacker may see:

- Usernames
- Passwords
- Cookies
- Session information



Protection Against Sniffing

Use encryption!

Examples:

- HTTPS (Encrypted websites)
- VPN (Virtual Private Network)
- WPA3 Wi-Fi Security
- SSH (Secure Shell)



Even if attacker captures packets, they cannot understand encrypted data.

Difference Between Bandwidth and Throughput

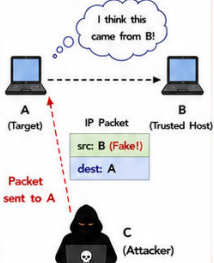
Concept	Meaning
Bandwidth	Maximum possible capacity of a link (theoretical limit)
Throughput	Actual achieved speed (real-world speed)

★ Bandwidth is how much a link **CAN** handle.
Throughput is how much it actually **DOES** handle.

2. Fake Identity — IP Spoofing

IP spoofing is an attack in which an attacker sends packets with a fake source IP address.

What Happens?



Why Is This Dangerous?

Attacker can:

- Pretend to be a trusted device
- Bypass filters and restrictions
- Hide real identity
- Launch attacks anonymously
- Steal data or perform malicious actions

⚠ It breaks trust in the network!

Important Note

IP spoofing changes the packet **header** (source address) — NOT the physical sender.



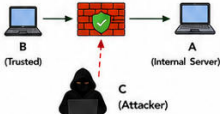
Like writing a fake return address on a letter.

Real-Life Example

Suppose a firewall trusts company computer B. Attacker C spoofs B's IP address and sends malicious packets.

Firewall thinks:

"This is from B, allow it."



Protection Against Spoofing

Defenses include:

- Packet filtering (firewalls)
- Authentication mechanisms
- HTTPS / TLS (adds verification)
- Ingress filtering by ISPs (BCP 38)
- Anti-spoofing / IDS systems

Key Takeaway

Attackers can lie about their identity at the IP level.

Always verify the source and use security controls.



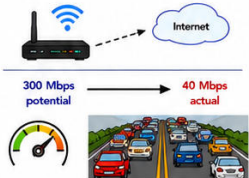
Example

Your Wi-Fi may support:

300 Mbps bandwidth

But actual throughput may be:

40 Mbps only



Big Picture: Throughput answers: How much useful data is really delivered per second? And the answer is controlled by: **the bottleneck link.**

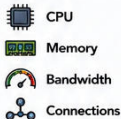


3. Denial of Service (DoS)

A Denial of Service (DoS) attack tries to make a server **unavailable** by overwhelming it with huge traffic from one or many sources.

What Is the Basic Idea?

A server has limited:

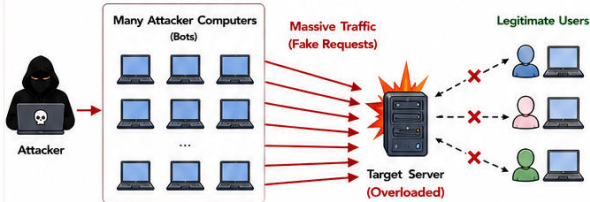


An attacker floods the server with fake requests.

Result:

Legitimate users cannot connect to the server.

How It Works (Example)



Analogy:



10000 people trying to enter one small shop at the same time – real customers cannot get in!

Three Steps of a DoS / DDoS Attack

Step 1 — Select Target

Attacker chooses a victim.

Examples:

- Website
- Game server
- Bank
- DNS server
- E-commerce site



Step 2 — Break Into Many Hosts

Attacker infects many computers worldwide with malware.

This group is called a **BOTNET**. Each infected computer becomes a **BOT**.



Step 3 — Attack Together

All infected machines (bots) send traffic to the target at the same time.

This creates a massive traffic flood!



Distributed DoS (DDoS)

When **MANY** computers attack together, it is called:

Distributed Denial of Service (DDoS)



DDoS is much stronger than a single-attacker DoS.

Real-World Examples

Famous services that have been hit by DDoS attacks:

- GitHub
- Twitter / X
- Netflix
- Major Banks
- Gaming Servers



Types of DoS Traffic

Attackers may send:

- Fake HTTP/HTTPS requests
- Huge file downloads
- SYN floods (many connection requests)
- Ping floods
- DNS amplification traffic
- UDP floods



Effects of DoS

Victims may experience:

- Slow website
- Timeout errors
- Connection failures
- Service unavailable
- Server crashes



Protection Against DoS / DDoS

Common defense techniques include:



Firewalls



Rate Limiting



Traffic Filtering



CDNs (Content Delivery Networks)



Load Balancing



Cloud DDoS Protection (e.g., Cloudflare, AWS Shield)

Why These Defenses Help

- ✓ Block malicious traffic
- ✓ Limit attack impact
- ✓ Keep service available
- ✓ Protect real users

Big Picture — Three Major Network Security Threats

Attack	Goal
Packet Sniffing	Steal information (eavesdrop on network traffic)
IP Spoofing	Fake identity (pretend to be someone else)
DoS / DDoS	Disable service (make server unavailable)

Simple Analogy — Real World

Network Attack	Real-World Analogy
Packet Sniffing	Secretly listening to someone's phone call.
IP Spoofing	Using a fake caller ID to pretend to be someone.
DoS / DDoS	Crowd blocking the entrance of a store so real customers cannot get inside.

Lines of Defense

Networks use multiple security defenses to protect against attacks like **sniffing, spoofing, malware, and DoS attacks**. These protections are called **security defenses**.

1. Authentication

Authentication means proving you are really who you claim to be.

Example (Gmail Login)

Username: user@gmail.com
 Password: *****
 OTP Code: 123456

Login

Server Identity Verified

Cellular Network Example

Mobile networks use **SIM cards**. SIM card contains unique **hardware identity** that helps the mobile provider verify the user.

Traditional Internet

On the normal Internet, there is no built-in **hardware identity**. Anyone can connect. That's why passwords and login systems are important.

Real-Life Analogy



Authentication is like showing your ID card at the airport.

2. Confidentiality

Confidentiality means keeping data secret using encryption.

What Is Encryption?

Encryption changes readable data into unreadable scrambled form. Only authorized users can decode it.



Why Important?

- Protects against:
- Packet sniffing
 - Hackers on Wi-Fi
 - Data theft



Examples of Encryption

Technology	Purpose
HTTPS	Secure websites (SSL/TLS)
VPN	Secure tunnel over Internet
WPA3	Secure Wi-Fi networks
TLS	Encrypted communication

Real-Life Analogy



Encryption is like putting a message inside a locked safe. Only the person with the key can open and read it.

3. Integrity Checks

Integrity means data was **NOT** modified during transmission.

The Problem

An attacker may change a message, inject fake data, or modify a transaction.



The Solution

Use digital signatures or hashes. They detect any tampering or changes.

Example

Original Message	Attacker Modifies
Transfer \$100	Transfer \$10000

Integrity checks detect the change and reject the message.

Real-Life Analogy



Integrity check is like a tamper-proof seal on a medicine bottle. If the seal is broken, someone changed it.

4. Access Restrictions

Access restrictions mean **only authorized users** may enter.

Example: Password-Protected VPN

A VPN creates a secure, private connection.

Only users with:

- Password
- Certificate
- Authorization can connect.



What Is VPN?

VPN = Virtual Private Network
 Creates an encrypted tunnel over the Internet.

- Used by:
- Companies
 - Universities
 - Remote workers



Real-Life Analogy



Access restriction is like a door lock with a key card. Only people with the card can open the door.

5. Firewalls

A firewall is a security device or software that **filters network traffic**.

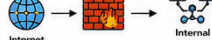
What Does Firewall Do?

Checks packets and decides: **ALLOW or BLOCK**



Middleboxes

Firewalls sit between networks like a security checkpoint.



Firewall Rules (Can block based on)



Off-by-Default

Block everything unless explicitly allowed. This is the safest policy.

Example

- Firewall may block:
- Hackers
 - Suspicious traffic
 - Unknown apps

but allow:

- Web browsing
- Email
- Business apps

Firewalls Against DoS

- Firewalls can:
- Detect huge traffic spikes
 - Rate limit attackers
 - Block malicious IPs
 - Help reduce DoS attacks

Real-Life Analogy



Firewall is like a security guard at a building entrance checking everyone before entry.

Big Picture — Main Defenses

Defense	Protects Against
Authentication	Fake identity, impersonation
Encryption	Packet sniffing, data theft
Integrity Checks	Data tampering, modification
Access Restrictions	Unauthorized access
Firewalls	Attacks, malicious traffic, DoS/DDoS

Complete Security Goals

Goal	Meaning
Authentication	Who are you?
Confidentiality	Keep data secret
Integrity	Prevent changes
Availability	Keep services running
All four goals together provide strong and trustworthy network security.	

Real-World Analogy Summary

Network Attack	Real-World Analogy
Packet Sniffing	Secretly listening to a phone call.
IP Spoofing	Using a fake caller ID.
DoS / DDoS	Crowd blocking the entrance of a store.



Important Insight:

Modern Internet security depends on **multiple layers of defense together**, because no single protection is enough.



Date: /

Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- **Protocol layers, service models**
- History



1. Protocol "Layers" and Service Models

Networks Are Very Complex Systems

A network contains many parts:

- Hosts (computers)
- Routers
- Cables / Wireless links
- Applications
- Protocols
- Hardware and software



Big Question

How can we organize all this complexity?

Idea of Layers

Instead of building one giant, complicated system, we divide networking into:

- ✓ Smaller organized parts
- ✓ Each part has its own job

Real-Life Analogy: Restaurant

	Customer	Orders food
	Waiter	Carries order
	Kitchen	Cooks food
	Delivery	Sends food

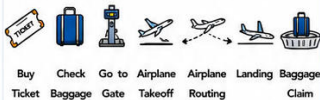


Each layer does **ONE** task. Layers work together to deliver the complete service.

2. Air Travel Example

End-to-End Travel

When you travel by air, you go through many services:



Important Idea

The entire trip works because multiple layers (services) cooperate to get you from start to finish.

Airline Layer Model (Services)

Service Layer	What It Does
Ticketing Service	Handles tickets and reservations
Baggage Service	Handles luggage
Gate Service	Handles loading and unloading
Runway Service	Handles takeoff and landing
Routing Service	Handles movement of airplane

Networking Similarity

Internet works the same way! Each networking layer provides a service to the layer above it.



Key Rule for Layers

- ✓ Each layer performs its own task
- ✓ Each layer hides internal details
- ✓ Each layer uses services of lower layers

3. Why Layering?

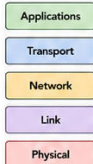
A. Easier Organization

Layers help us understand complex systems by breaking them into smaller pieces.

Example:

You can study:

- Wi-Fi separately
 - Routing separately
 - Applications separately
- ... without learning the entire Internet at once.



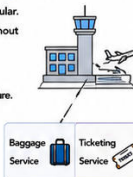
B. Modularization

Layers make systems modular. One layer can change without affecting others.

Real-Life Example:

Airport changes gate procedure. But:

- Airplanes still fly
- Baggage still works
- Tickets still work



Networking Example:

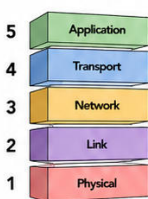
Wi-Fi can change from Wi-Fi 5 → Wi-Fi 6 without changing:

- Websites
- Email
- YouTube



4. Layered Internet Protocol Stack

The Internet is organized into stack of **5 layers**.



Internet Layers	Main Job
5 Application Layer	Supports user applications
4 Transport Layer	Provides process-to-process delivery
3 Network Layer	Routes packets between hosts
2 Link Layer	Delivers data on the local link
1 Physical Layer	Sends raw bits over the medium

1. Application Layer

Supports applications that users interact with.

Examples:

- HTTP → websites
- SMTP → email
- DNS → domain names
- IMAP → mail retrieval



What Happens Here?

Applications create messages. Examples: web request, email, chat message, etc.



2. Transport Layer

Responsible for process-to-process communication.

Main Protocols:

- | | |
|--|--|
| TCP <ul style="list-style-type: none"> • Reliable • Ordered • Error recovery | UDP <ul style="list-style-type: none"> • Faster • Less reliable |
|--|--|

Used in:

- Websites
- File transfer
- Email

TCP = reliability
UDP = speed (less reliability)

3. Network Layer

Responsible for routing packets across networks.

Main Protocol:

IP (Internet Protocol)

Routers mainly work here.



Big Picture: The Internet works because each layer provides a service to the layer above it, using the services of the layer below it.



Multiple layers cooperate to deliver communication end to end!

4. LINK LAYER

Responsible for:
communication between
neighboring devices.

Examples:

- Ethernet (802.3)
- Wi-Fi (802.11)
- PPP (Point-to-Point Protocol)

Example:

Laptop → Wi-Fi Router



This communication is
handled by: **LINK LAYER**

5. PHYSICAL LAYER

Lowest layer.

Responsible for sending raw bits
through:

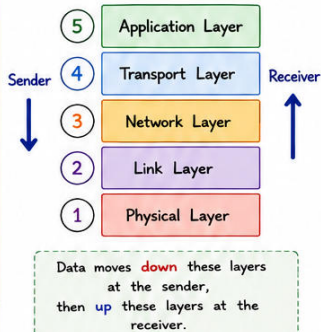
- Electrical signals (copper cables)
- Radio waves (wireless)
- Light pulses (fiber optics)

Example:

Binary data: **10101010**
travels physically through
cables or air.



COMPLETE STACK



5. SERVICES, LAYERING, AND ENCAPSULATION

1 Application Creates Message

Application creates a message: **M**

Examples:

- Webpage request
- Email text
- Video data



2 Transport Layer Encapsulation

Transport layer adds a transport header
called: **Ht**

Result: **Ht M**

This is called: **SEGMENT**

What is inside Transport Header?

Contains information like:

- Port numbers
- Sequence numbers
- Error checking

3 Network Layer Encapsulation

Network layer adds a network header
called: **Hn**

Result: **Hn Ht M**

This is called: **DATAGRAM**

What is inside Network Header?

Contains information like:

- Source IP address
- Destination IP address
- Routing information

4 Link Layer Encapsulation

Link layer adds a link header
called: **Hi**

Result: **Hi Hn Ht M**

This is called: **FRAME**

What is inside Link Header?

Contains information like:

- MAC addresses (source & dest)
- Error detection (FCS)
- Frame control info

Encapsulation Meaning

Encapsulation means wrapping
data with headers at every layer.
Like packing boxes inside
bigger boxes.



Real-Life Analogy: Shipping a Gift

Step	Added	Looks Like
1	Gift (original message)	
2	Small box (transport header)	
3	Shipping box (network header)	
4	Shipping label (link header)	

At Receiver Side (Decapsulation)

Receiver removes headers in reverse order:

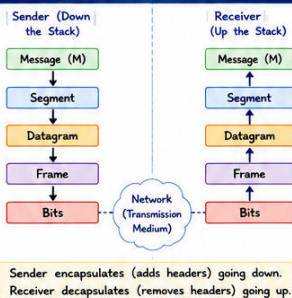
- 1 Remove link header (Hi)
- 2 Remove network header (Hn)
- 3 Remove transport header (Ht)
- 4 Get original message (M)

This process is called: **DECAPSULATION**

Names of Data at Each Layer

Layer	Data Name
5 Application Layer	Message (M)
4 Transport Layer	Segment
3 Network Layer	Datagram / Packet
2 Link Layer	Frame
1 Physical Layer	Bits

Full Communication Flow



Most Important Concepts Here

Concept	Meaning
Layering	Divide networking into organized parts
Service	Each layer provides functionality to the layer above
Encapsulation	Adding headers layer by layer
Modularity	One layer can change without affecting others
Protocol Stack	The collection of Internet layers that work together



Big Picture:

The Internet works because each layer provides a service to the layer above it, using the services of the layer below it.



Multiple layers cooperate
to deliver communication
end to end!

1. Encapsulation = Matryoshka Dolls

Main Idea

Each networking layer wraps the previous layer with extra information called: **headers**

Exactly like:

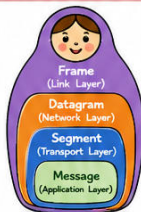
- a small doll goes inside bigger doll
- then inside even bigger doll



Networking Version

Layer	Data Name
Application	Message
Transport	Segment
Network	Datagram
Link	Frame

Final Encapsulation Stack (Like Matryoshka Dolls)



Important Observation

Every lower layer encapsulates the data from above.

Step-by-Step Encapsulation (Wrapping Data with Headers)

1 Application Layer

Application creates: **message (M)**

Example:

- webpage request
- email
- chat message



Result: **M**

2 Transport Layer

Transport layer adds: transport header (Ht)

Result:



This becomes: **segment**

What Does Transport Header Contain?

Examples:

- port numbers
- sequence numbers
- reliability information



3 Network Layer

Network layer adds: network header (Hn)

Result:



This becomes: **datagram**

What Is Inside Network Header?

Contains:

- source IP
- destination IP
- routing information



4 Link Layer

Link layer adds: link header (Hl)

Result:



This becomes: **frame**

What Is Inside Link Header?

Contains:

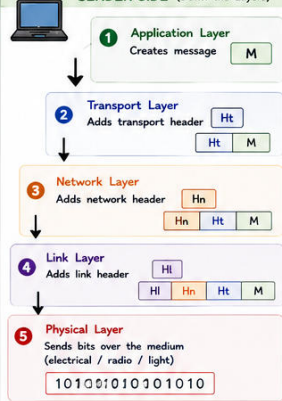
- MAC addresses (source & dest)
- error detection (FCS)
- frame control info



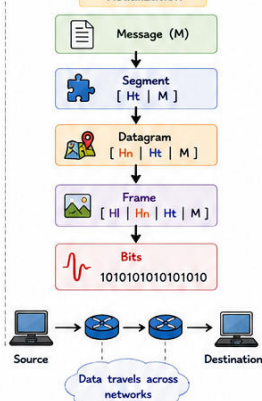
2. Full Sender → Receiver Process

Complete communication between source and destination using encapsulation (down) and decapsulation (up).

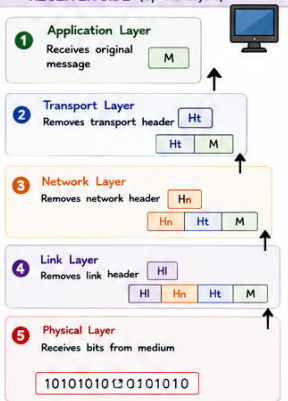
SENDER SIDE (Down the Layers)



Visualization



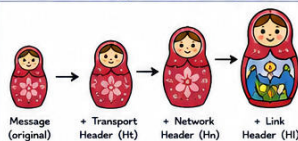
RECEIVER SIDE (Up the Layers)



Names of Data at Each Layer

Layer	Data Name
Application	Message
Transport	Segment
Network	Datagram / Packet
Link	Frame
Physical	Bits

Real-Life Analogy: Russian Nesting Dolls



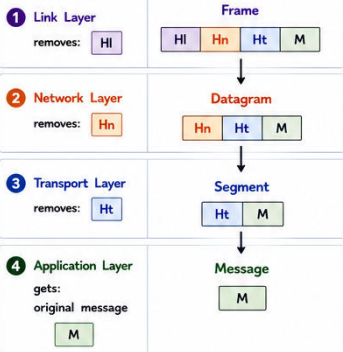
Key Takeaways

- ✓ Encapsulation: headers are added as data moves down the stack.
- ✓ Decapsulation: headers are removed as data moves up the stack.
- ✓ Each layer provides a service to the layer above.
- ✓ Lower layers handle details; upper layers focus on the task.
- ✓ This layering makes complex networks possible!

1. DECAPSULATION (At the Receiver)

Decapsulation = Receiver removes headers step-by-step.

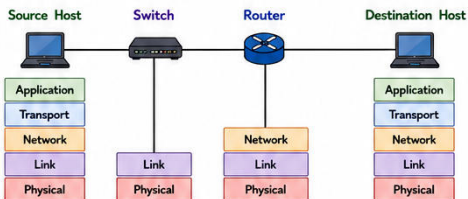
Step-by-Step



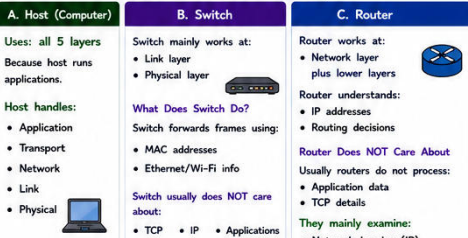
This Is Called: **DECAPSULATION**
Meaning: **unwrapping** the layers

2. ENCAPSULATION: END-TO-END VIEW

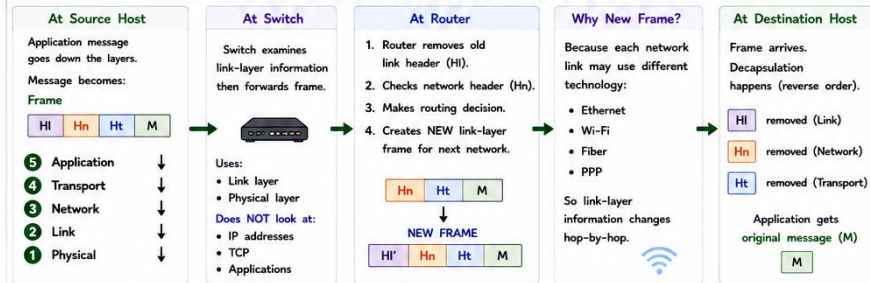
This is one of the **MOST IMPORTANT** networking diagrams.
It shows: hosts, switches, routers, layers used by each device.



Different Devices Use Different Layers



Important Flow: How Data Moves Across the Network



Key Insight

What Stays End-to-End?

Usually these travel across the network:

- Application message (M)
- Transport header (Ht)
- Network header (Hn)

What Changes Hop-by-Hop?

Link-layer frame (HI) changes at every router/switch.

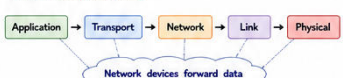
Simple Analogy: Package Shipping

Networking	Real Life
Message (M)	Actual gift
Transport Header (Ht)	Tracking info (who sent)
Network Header (Hn)	Destination address
Link Frame (HI)	Local delivery packaging (changes at each city)

At each city:
Outer local packaging changes, but destination address remains the same.

Full Communication Summary

Sender (Down the Stack)



Receiver (Up the Stack)



Most Important Concepts From These Slides

Concept	Meaning	Concept	Meaning	Concept	Meaning
Encapsulation	Wrapping data with headers	Segment	Transport-layer unit (after adding Ht)	Switch	Works mainly at link layer (uses MAC addresses)
Decapsulation	Removing headers	Datagram	Network-layer unit (after adding Hn)	Router	Works mainly at network layer (uses IP addresses)
Message	Application-layer data	Frame	Link-layer unit (after adding HI)	Host	Uses all 5 layers

Date: /